NameRank: Measuring LLM-Mediated Recognition as the Post-Bibliometric Impact Channel

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Abstract

What a frontier language model says about a researcher, founder, or tool is increasingly the first description a curious human sees—so whether an entity's name has propagated into the models' training corpora is now a measurement problem in its own right. Bibliometrics answer only a fraction of it: citation counts explain about a third of the variance in whether a model recognizes a researcher [Li, 2026]. We treat the residual as the quantity of interest and introduce **NameRank**, a continuous $[0,1]$ recognition score: each of 5,719 entities in 54 cohorts is probed with one open-ended question across 37 frontier models, and an independent LLM judge scores every response against a curated gold answer on multiplicative coverage$\times$accuracy, so fluent hallucinations earn nothing (211,603 records, plus a Chinese-prompt sub-run). Six findings stand out. **(i) The credential treadmill:** seven of nine Olympic-style credentials (IMO, IOI, CMO, Rhodes, MSRA) score at or below a long-tail working-researcher baseline—the credential no longer propagates a name; named post-credential production does. **(ii) The artifact-over-creator inversion:** in 8 of 11 verified (creator, artifact) pairs the tool out-ranks its maker, and a context-injection experiment corroborates the attribution mechanism. **(iii) Among bibliometrics, h-index carries the signal:** citations add nothing once it is in the model, because NameRank rewards name-recurrence across distinct works—yet even h-index leaves $\sim$60% of recognition variance unexplained. **(iv) A corpus-density gradient:** Stanford-affiliated CS faculty score roughly twice Tsinghua-affiliated, part of an English-corpus ladder that runs across institutions, countries, and prompt languages. **(v) An external calibration on news events:** on 258 events of 2021–2023, whose worldwide attention is directly recorded as a daily pageview ledger, recognition tracks attention dose but loads on an event's *peak* salience rather than its persistence—recognition follows document production, not audience retention—and year-templated series names (elections, tournaments) are penalized at matched attention. **(vi) Self-report is not measurement:** asked to pairwise-rank their own recognition, frontier models reconstruct the panel ordering only partially ($\rho \approx 0.5$–0.6) and read shared corpus salience rather than their own knowledge state—yet they almost never claim a fictional name, erring by under-claiming. Below its threshold the metric is silent rather than misleading, and the findings survive paraphrase, training-cutoff, cross-judge, synthetic-null, and confound controls. We release all probes, gold answers, responses, and code.

Code: `https://github.com/19PINE-AI/namerank`
Website: `https://01.me/research/namerank`

1 Introduction

The channel through which human curiosity reaches people and their work is changing. A practitioner deciding whether to read a paper, hire a candidate, attend a talk, or invest in a startup increasingly arrives at the question with whatever a frontier language model says about the entity, often before any human source intervenes. We do not measure the prevalence of this LLM-first lookup behavior—it is a usage trend we take as a premise, not a result—but conditional on it, what the model says is determined by whether the entity's name has propagated into the model's training corpus. If the corpus has absorbed the name with a clean attribution chain, the practitioner is presented with a coherent summary; if it has not, the model says "unknown" or, worse, fabricates a plausible-sounding biography. The information surface that mattered ten years ago—search rankings, conference plenaries, citation counts—has been re-mediated by a small number of frontier models, and whether a name has propagated into those models' weights is now itself a substantive measurement problem.

Bibliometrics do not solve this measurement problem. In earlier work on the Incompressible Knowledge Probes (IKP) benchmark, Li [2026] found—as a byproduct of calibrating the knowledge capacity of frontier models—that citation counts explain only about a third of the variance in whether a model recognizes a researcher. A controlled audit attributed the remainder to three corpus-level mechanisms: *name uniqueness*, *named-artifact attachment*, and *subfield-ecosystem density*. That work characterized the residual but did not measure it: its recognition scores live on a coarse tier ladder built for parameter estimation, not for comparing entities. We take the opposite stance—the residual is the quantity of interest—and build a dedicated instrument for it. Section 6 details how the instrument departs from its predecessor.

A heuristic factorization makes the target precise (a framing device, not an identity we estimate—the second factor is left unmeasured):

$$\text{Recognition impact} \approx \underbrace{\text{NameRank}}_{\text{per-query presence}} \times \underbrace{\text{Query volume}}_{\text{humans querying that LLM}}$$

To the extent the query volume is approximately constant across entities at the relevant time horizon, the first factor carries the cross-entity variation, and it is what we measure. We do not claim NameRank measures *quality* or *merit*; it measures whether the corpora that mediate human curiosity at scale have internalized the entity's name.

NameRank is a continuous $[0, 1]$ recognition score. Each entity is probed with one fixed open-ended question ("tell me what you know about [name], who [context]") across a panel of 37 frontier models. An independent LLM judge scores every response against a curated gold answer on separate coverage and accuracy axes, and the two are multiplied, so that a fluent hallucination—high topical coverage, wrong specifics—contributes nothing. The per-entity score is the panel mean. Three properties distinguish this construction from recognition numbers that fall out of benchmark suites: it places researchers, founders, open-source maintainers, and named artifacts on a single axis; it is hallucination-resistant by construction; and cross-model disagreement is retained as a signal rather than averaged away. We evaluate 5,719 entities across 54 cohorts spanning academia, industry, open source, and the public sphere (Section 2).

Contributions.

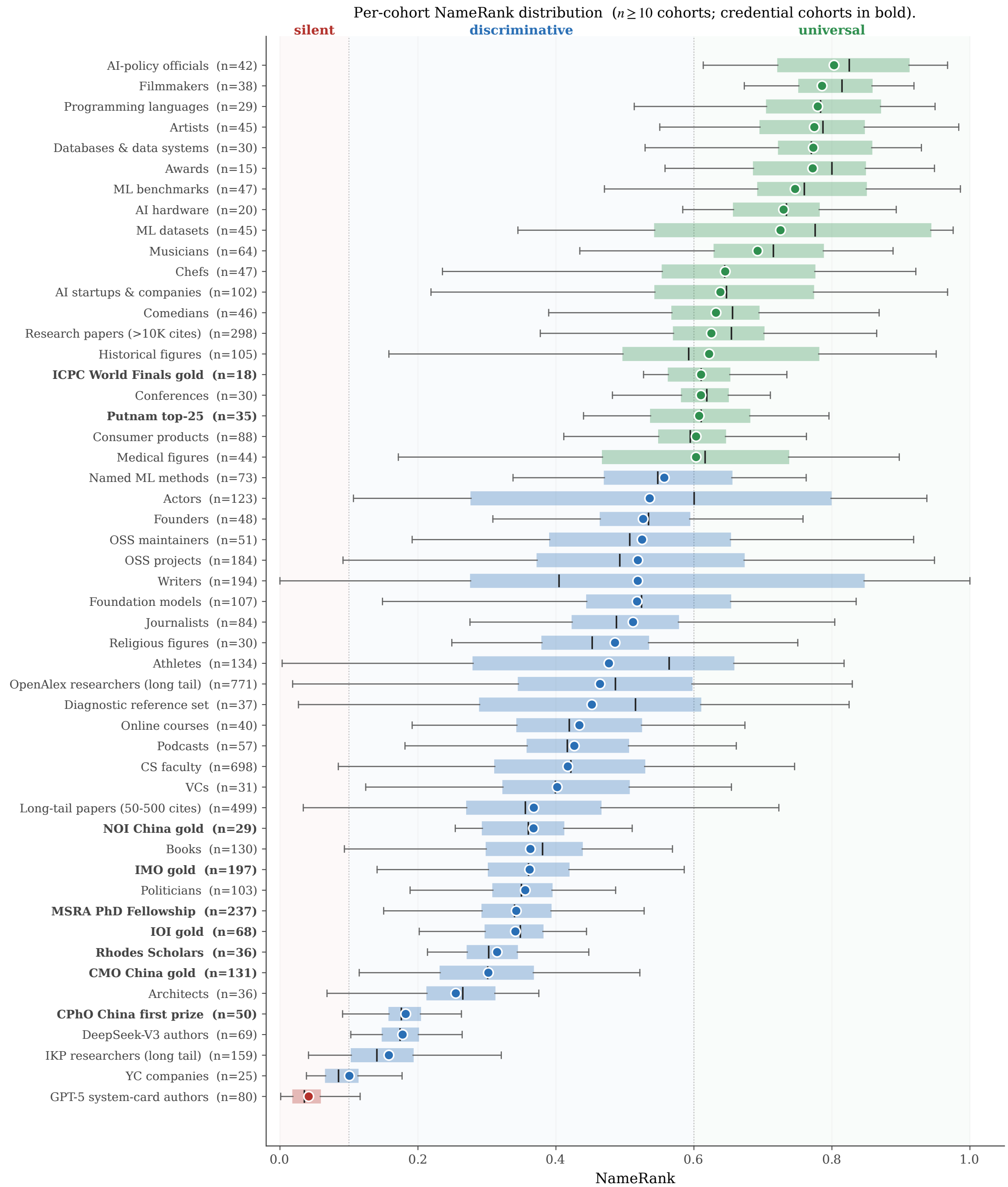

Figure 1. Per-cohort NameRank distribution. Each box is a cohort ($n \geq 10$); the dot is the cohort mean, the box covers the IQR, and the inner line is the median. Three zones are visible on the same axis: *silent* (mean ≤ 0.1), *discriminative* (0.3–0.6: working academics and credentialed individuals), and *universal* (> 0.7: named artifacts and high-profile mid-tier figures). The nine credential cohorts (bold labels) cluster in the lower half of the discriminative zone (Section 3.2).

1. A **continuous cross-domain recognition instrument** that places researchers, founders, OSS authors, public websites, and named artifacts on one $[0,1]$ axis—no prior instrument does (Section 3.1).

2. The **credential treadmill**: most Olympic-style intellectual credentials (IMO, IOI, CMO, Rhodes, MSRA Fellowship) score at or below a working-researcher baseline. The credential no longer propagates a name; named post-credential production does (Section 3.2).

3. The **artifact-over-creator inversion**: for independent creators, the tool out-ranks its maker. The effect is measured observationally, corroborated by a context-injection experiment, and traced to an attribution-direction asymmetry in the corpus (Section 3.3).

4. **What predicts recognition**: among bibliometrics, h-index carries the signal—NameRank rewards name-recurrence across distinct works rather than citation volume—though even it explains under half the variance; a corpus-density gradient runs across institutions, countries, and prompt languages; and a 258-event news cohort with a directly recorded attention ledger calibrates the instrument externally—peak salience, not persistence, drives propagation (Sections 3.4–3.5).

5. A **validation battery** covering wording, model vintage, judge family, synthetic nulls, and gold-answer confounds (Section 4); a **self-report probe** showing that pairwise recognition introspection reads the corpus prior and cannot replace verified measurement (Section 3.7); and an **open release** of all probes, gold answers, responses, and code (links on page 1).

Figure 1 previews the central result: a single recognition axis on which every cohort in the study is placed.

2 Measuring NameRank

Figure 2 summarizes the pipeline this section unpacks: a fixed open-ended probe is sent to a 37-model panel for each entity; an independent LLM judge scores every (entity, model) response against a curated gold answer on multiplicative coverage×accuracy; and the per-entity NameRank is the unweighted panel mean.

2.1 Definition

For an entity e with a disambiguating context $c(e)$ and a curated gold answer $g(e)$, and a model panel $\mathcal{M}$ of size $M = 37$, NameRank is

$$\mathrm{NameRank}(e) \;=\; \frac{1}{M}\sum_{m\in\mathcal{M}} \mathrm{cov}(r_{e,m}, g(e)) \cdot \mathrm{acc}(r_{e,m}, g(e)), \tag{1}$$

where $r_{e,m}$ is the response of model m to the open-ended probe

> *"Tell me what you know about [name], who [context]. If you do not recognize this entity, respond with 'unknown'. Limit your response to about 150 words."*

and $\mathrm{cov}, \mathrm{acc} \in [0,1]$ are independent coverage and accuracy scores produced by an LLM judge (Section 2.4). The multiplication is critical: a fluent-hallucination response with high topical coverage but factually wrong specifics has $\mathrm{cov} \to 1$ but $\mathrm{acc} \to 0$, and contributes nothing. Three worked (entity, model, response) triples spanning the score range are provided in the repository documentation.

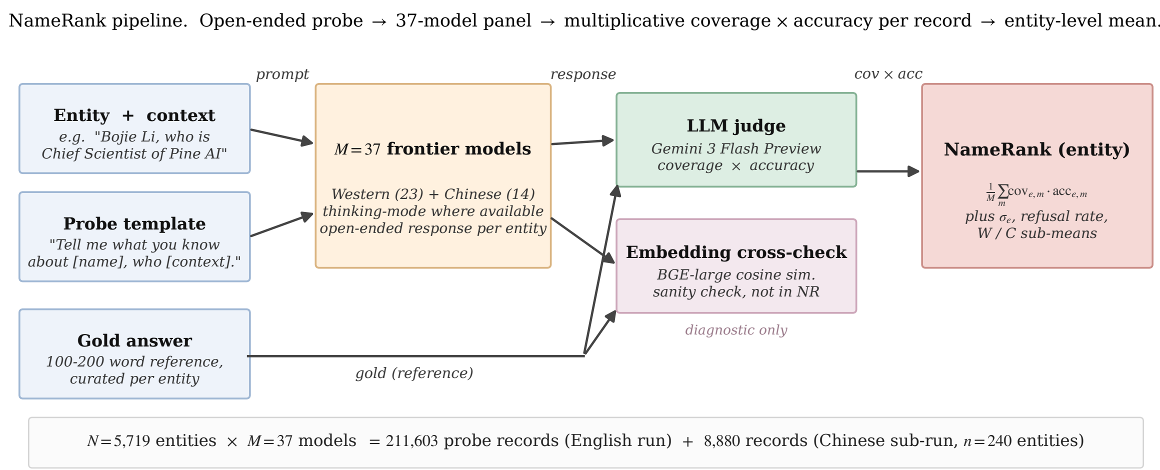

Figure 2. NameRank pipeline. The probe template is fixed, with two slots: the entity name and a short role/affiliation/field context. Each of the 37 panel models answers the open-ended probe; an LLM judge grades the response against the gold answer on independent coverage and accuracy axes in $[0, 1]$, and their product is the per-record contribution. An embedding similarity (BAAI/bge-large-en-v1.5) is stored for diagnostics only—it is *not* part of the score (Appendix K). Totals: 211,603 English-prompt records over 5,719 entities, plus an 8,880-record Chinese-prompt sub-run on a 240-entity subset.

2.2 Probe Design

One open-ended probe per entity. Closed-factoid probes have a known failure mode: a model that recognizes the entity through fact X but does not recall fact Y scores zero on a probe-of-Y, biasing the metric toward specific-fact recall. The open-ended probe instead asks the holistic question: does the model have a retrievable representation of this entity? The template is fixed, with two slots: the canonical entity name, and a short disambiguating clause giving role, affiliation, and field (e.g., *"Wei Zhang, who is a professor of mathematics at MIT working on PDEs"*). The context resolves name collisions at probe time without revealing the gold answer. The full English and Chinese probe text and the judge prompt are reproduced verbatim in the repository documentation.

The context names role, affiliation, and field—never specific contributions, papers, dates, or named artifacts—so it anchors disambiguation without leaking the gold; the judge's rubric credits only specific named facts, and generic in-context biographies earn nothing (evidence in Appendix A; a dedicated context-ablation in Appendix P).

2.3 Gold Answers

For each entity we construct a 100–200 word gold answer covering notable contributions, affiliations, and identifying facts, sourced from Wikipedia intros (∼60% of the cohort), OSS READMEs (∼10%), company pages and news coverage (∼10%), and manual curation from personal sites, competition archives, and OpenAlex profiles for the long tail (∼20%). Gold-answer quality matters less than one might expect, because the judge's coverage score measures which of the gold's named facts appear in the response: an omission in the gold affects scale but not within-cohort ranking. Because most golds derive from Wikipedia, coverage partly measures whether the model absorbed the same text that seeds the gold; the overlap is intended—Wikipedia presence *is* part of corpus presence—and a dedicated check confirms NameRank does not reduce to a has-a-Wikipedia-page flag (Appendix P). We audit a stratified 5% sample against primary sources, following the audit protocol of Li [2026]; the observed correction rate is ∼3%, typically year-of-affiliation drift.

2.4 The Judge

We use Gemini 3 Flash Preview as the judge (direct Google API to minimize routing variance; temperature 0; low reasoning budget). The judge prompt presents the gold and the response and elicits two independent scores in $[0, 1]$:

- **Coverage**: how much of the gold's *substantive* content appears in the response, where substantive means named artifacts, named affiliations, named contributions, and identifying facts (year, location, role title). Vague statements like *"works in AI"* earn no credit.
- **Accuracy**: whether the factual claims in the response are correct per the gold, a factual claim being a specific named entity, year, role, or relationship.

The prompt carries an explicit hallucination instruction (*"Hallucinated bios—high coverage, low accuracy—MUST score LOW on accuracy"*). Refusals score 0 on both axes; correct extra facts absent from the gold are not penalized; vague statements that could apply to many entities are not credited.

Alongside the judge we record, for every record, the embedding cosine similarity between response and gold [Xiao et al., 2024]. It is a diagnostic only and never enters the score: as Appendix K shows, embedding similarity systematically credits fluent hallucinations as recognized. The dependence of the findings on the judge's model family is quantified with a three-judge re-grade in Appendix N.

2.5 Aggregation

NameRank is the unweighted mean of cov · acc across the 37-model panel. We also retain the within-entity standard deviation across models, the refusal rate, the Western/Chinese sub-panel means, and the raw response text. Averaging across the panel is what makes the score stable: a single model contributes noise from prompt sensitivity, refusal-policy idiosyncrasies, and training-data accidents, while the panel mean integrates corpus presence over the current frontier fleet. The retained per-model scores keep the residual signal—an entity recognized by one vendor's models but not another's is itself informative (Section 3.4.3).

2.6 Model Panel

The panel comprises 37 frontier and near-frontier models accessed via OpenRouter—23 Western and 14 Chinese, spanning frontier-reasoning to 1B-parameter classes (full roster in Appendix A, Table 5). Thinking mode is enabled wherever a thinking variant exists: for a recognition metric, thinking mode is the operational mode in which most user queries are answered, so the panel reflects deployment reality. The Western/Chinese partition is used only for the cross-vendor analysis (Section 3.4.3); the headline NameRank is the unweighted mean over all 37 models.

2.7 Entity Cohorts

We evaluate $N = 5{,}719$ entities in 54 cohorts, spanning seven groups: credentialed individuals (801: olympiad medalists, Putnam, ICPC, Rhodes, MSRA fellows), working academics (1,628), industry figures (248), named artifacts (1,332), mid-tier cross-profession figures (1,461: writers, athletes, actors, chefs, podcasts), a hand-curated 37-entity diagnostic reference set with higher-detail golds (public figures, engineers, and artifacts used for case-level analyses), and special cohorts (212: GPT-5 system-card authors, DeepSeek-V3 authors, conferences, awards). The design goal is coverage of the axes the metric claims to unify: credentialed individuals whose recognition should be carried by the credential if credentials propagate; working academics with known bibliometrics, as the external-validity anchor; industry figures whom no bibliometric measures; named artifacts, to compare things against their makers; and mid-tier figures, to test the axis outside the technology world. The per-group breakdown is in Appendix A (Table 6); full per-cohort sizes, means, and inclusion criteria are in the released cohort specification.

2.8 Cross-Language Sub-Run, Execution, and Release

A 240-entity subset—the diagnostic reference set, the Chinese olympiad and fellowship cohorts, the DeepSeek-V3 author cohort, and small Western control samples—was re-probed with the Chinese-translated template (8,880 additional records; gold answers and judge prompt remain English, since the judge grades across languages; cohort selection detail in Appendix H). A parallel Python runner completed the full English run in ∼10 hours of wall-clock time and the Chinese sub-run in ∼30 minutes, at an end-to-end API cost of ∼$2,500 including iteration. We release the probe specifications, all 5,719 gold answers, all judged response records, the Chinese sub-run, and the analysis pipeline at `https://github.com/19PINE-AI/namerank` (companion website: `https://01.me/research/namerank`).

2.9 Scope: What NameRank Does and Does Not Measure

NameRank is not a quality, merit, or intrinsic-impact metric. It measures *the propagation of a specific name into the indexed corpora that frontier-model training pipelines scrape*. This correlates with academic impact through bibliometric mechanisms, with cultural reach through named-artifact attachment, and with both through derivative-content density (tutorials, blog posts, talks, podcast mentions)—but it does not measure the underlying work. The clearest counterexample in our data is a Chinese-language visa-monitoring website with roughly a million users whose NameRank is nonetheless low, because its name never co-travels with its creator's in indexable text (Section 3.6). The metric correctly registers low there, not because the artifact lacks impact, but because the metric measures something specific. We treat this as a feature: a clean instrument reports what it measures, not what users wish it measured.

Three empirical properties, each established later, govern how the score should be read. First, the axis is meaningful *across* domains: researchers, founders, OSS authors, and artifacts land on one scale (Section 3.1). Second, below the corpus-presence threshold the metric is largely *silent rather than misleading*—the panel refuses instead of producing spurious rankings—with a noise floor that depends on how specific the gold answer is (Section 3.1, Appendix O). Third, the score directly captures the residual that bibliometrics leave unexplained: for a working researcher, one widely-used, cleanly-named open-source tool moves NameRank more than one well-cited paper (Sections 3.3, 3.4).

3 Results

3.1 A Single Axis Across Domains

Figure 1 shows the per-cohort NameRank distribution. The instrument resolves into three zones on one axis: *silent* (cohort means below 0.1: names that have not propagated, such as authors known only from a recent contributor list), *discriminative* (roughly 0.3–0.6: working academics and credentialed individuals, where the metric separates entities), and *universal* (above 0.7: named artifacts and high-profile figures that essentially every model knows).

The cross-domain placement is the headline result. Table 1 puts a best-selling author, AI executives, working researchers, open-source tools, a public-utility website, and a Fortune-10 CTO on the same scale. No prior *verification-based* instrument supports this placement: citation counts do not measure Sam Altman or a Walmart CTO, *h*-index does not measure the creator of nanoGPT, and GitHub stars do not measure Yuval Noah Harari. Web-attention metrics—search volume, Google Trends, Wikipedia pageviews—do place all of them on one axis, but that axis measures the flow of current curiosity, not the stock of verified knowledge, and it is empirically a different axis: article pageviews are undefined for the 71% of our entities with no Wikipedia article, explain little of NameRank where they are defined, and anticorrelate with it across technical cohorts (Appendix P, Figure 13).

Table 1. Selected cross-domain entities on a single NameRank axis, spanning the universal, discriminative, and silent zones. Diagnostic reference-set entities (Altman, Hinton, Karpathy, and below) carry deliberately high-detail gold answers, which cap coverage for any 150-word response—levels are gold-conditional (Section 3.5, Appendix O), so even universally known figures sit near 0.6 here. The same person can therefore carry different levels under different golds: the first author scores 0.09 here, against a dense biography gold, and 0.35 in Appendix C, against the short medalist-cohort gold—the gap is the denominator, not the models.

Entity	NameRank	Role
Yuval Noah Harari	1.00	best-selling author
Aravind Srinivas	0.66	CEO, Perplexity
Geoffrey Hinton	0.62	deep-learning pioneer
Sam Altman	0.61	CEO, OpenAI
Andrej Karpathy	0.61	founder, Eureka Labs
Tri Dao	0.53	FlashAttention author; Princeton faculty
Suresh Kumar	0.52	CTO, Walmart
Tianshou	0.42	open-source RL library
Jiayi Weng	0.33	Tianshou author; OpenAI engineer
tuixue.online	0.25	visa-monitoring website, $\sim$1M users
Bojie Li	0.09	this paper's first author
GPT-5 system-card authors	0.04	cohort mean over 80 contributors

Below the corpus-presence threshold, the metric goes quiet rather than wrong. The GPT-5 system-card author cohort—80 real, professionally accomplished people—averages 0.042, with 71% of panel responses returning "unknown" rather than a fabricated biography. Section 4 calibrates this silence against a synthetic-null floor and shows it is intrinsic rather than a corpus-timing artifact.

3.2 The Credential Treadmill

We measure nine prestigious competition and fellowship credentials—each once a career-defining distinction—against a baseline cohort of ordinary citation-tracked researchers (OpenAlex, 500–30,000 citations). Figure 3 shows the ladder; the full table, with cohort sizes and temporal breakdowns, is in Appendix B. Seven of the nine credential cohorts sit *at or below* the working-researcher baseline. An IMO gold medal—the canonical intellectual credential of the past four decades—predicts less LLM recognition than being an average citation-tracked researcher (0.36 vs. 0.46).

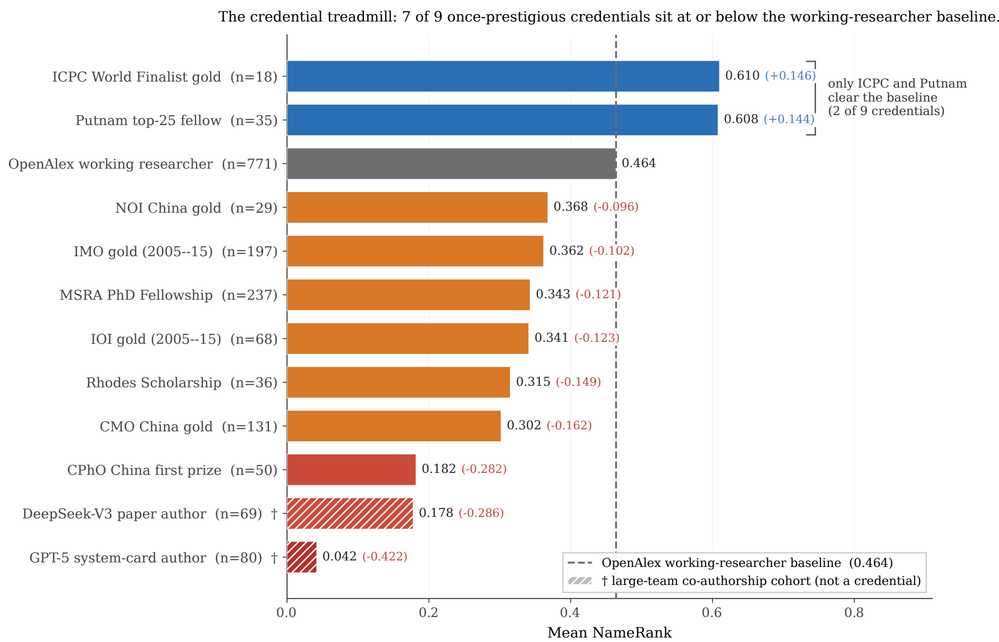

Figure 3. The credential treadmill. Bars are cohort mean NameRank; the dashed grey line is the working-researcher baseline (OpenAlex long-tail cohort, 0.464). Red: silent or near-silent cohorts. Orange: at or just below the baseline. Blue: above the baseline. Of the nine prestigious credentials, only two (ICPC, Putnam) clear the baseline. The two hatched bars (†) are large-team co-authorship cohorts shown for context, not credentials, and are excluded from the "7 of 9" count. Full table in Appendix B.

The two exceptions prove the mechanism. ICPC World Finalists and Putnam fellows clear the baseline—and both credentials feed almost exclusively into US-tracked academic and industry careers with high English-corpus density. They are not exceptions to credential decay; they are proxies for selection into careers that produce English text. IMO and CMO medalists, selected by the same kind of intellectual filter, disperse globally into careers whose post-medal output is often not in English, so the medal-and-name pair never co-occurs at density in the training corpus. Conditioning on career track confirms this: IMO golds on US-tracked careers score near 0.50, non-Anglophone-tracked golds near 0.30 (Appendix B). A complete-roster extension sharpens the same point from the other side: on all 103 medalists of a single contest (NOI 2009), recognition is a gold/non-gold cliff — the silver and bronze cohorts sit at the synthetic-null floor, within-tier contest score adds almost nothing, and the non-golds who are recognized owe it to post-medal careers, not the medal (Appendix C).

The comparison is deliberately conservative: the baseline is citation-floored while the credential cohorts are not, yet the lowest credential cohorts sit at the synthetic-null noise floor in absolute terms, and restricting IMO golds to citation-tracked careers sharpens rather than reverses the gradient (Appendix B).

The generational claim. Olympic-style credential systems were optimized for a pre-LLM signaling regime, in which a credential translated into reputation through human social networks. In the LLM-mediated regime, only credentials that translate into *named, indexable production*—papers, tools, blog posts, podcasts carrying the holder's name—propagate. The credential itself, absent subsequent production, does not.

3.3 Named-Artifact Amplification

3.3.1 The artifact-over-creator inversion

For 11 verified (creator, artifact) pairs we measure both members independently (Figure 4a; full table in Appendix D). In 8 of the 11 pairs, the artifact out-ranks its creator.[1] The three exceptions are senior leaders of large, young organizations (DeepMind, Thinking Machines Lab, Perplexity), where the leader accumulated independent corpus presence before the organization existed. For independent creators of single artifacts—Datasette, nanoGPT, Tianshou, LangChain, Cursor, FlashAttention—*the artifact uniformly exceeds the creator*.

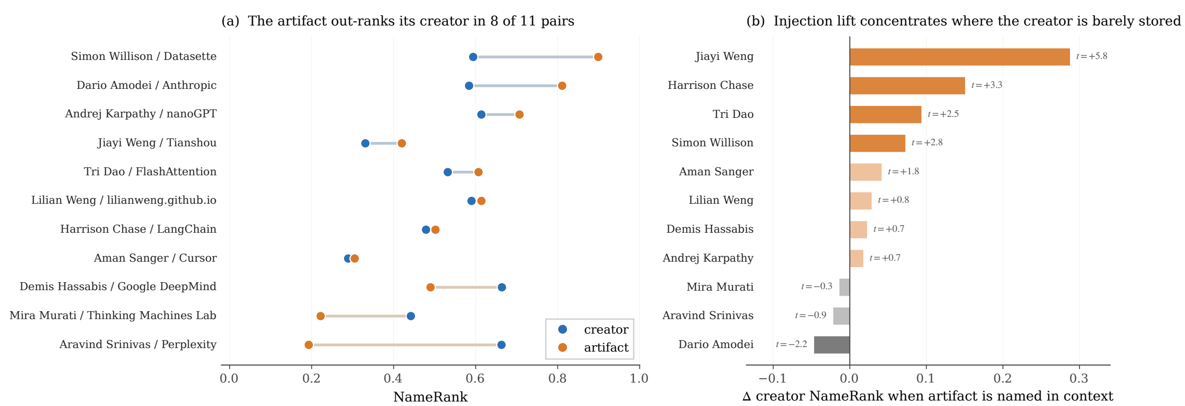

Figure 4. Named-artifact amplification. **(a)** Creator vs. artifact NameRank for the 11 verified pairs, sorted by artifact−creator delta: for independent creators of single artifacts the tool uniformly exceeds its maker; the three exceptions are senior leaders whose recognition predates the organization. **(b)** The context-injection experiment: adding one clause naming the artifact to the creator's disambiguating context lifts creator NameRank by $+0.058$ on average, with the lift concentrated in creators the panel barely stores (solid bars: $|t| \geq 2$ across the model panel). Full tables in Appendix D.

[1] Artifact rows use the bare product name as it circulates in text (*Perplexity*, *Cursor*). The release also contains disambiguated variants—*Perplexity AI (company)* scores 0.776 and *Cursor (company)* 0.840, far above the bare names (0.193, 0.305)—because qualification removes the collision with the perplexity metric and the UI cursor. The bare-name choice is conservative for the inversion count: it depresses the artifact side, and under the qualified variants the Srinivas/Perplexity exception itself flips, making 9 of 11 pairs invert. The collision penalty on bare artifact names is not a nuisance term; it is the name-uniqueness mechanism (Sections 3.5, 6) acting on artifacts.

3.3.2 Attribution flows from creator to artifact, not back

Measuring, for each pair, how often a probe about one member names the other (the $C \to A$ and $A \to C$ mention rates of Table 9, Appendix D) reveals three regimes. In the most common one—*famous artifact, anonymous creator* (Sanger/Cursor, Chase/LangChain, Srinivas/Perplexity, Amodei/Anthropic)—a probe about the creator reliably names the artifact, but a probe about the artifact frequently fails to name the creator. A second regime—*paired in corpus* (Lilian Weng and her blog, Tri Dao and FlashAttention)—shows the academic-norm pattern: creator and artifact co-occur in both directions. The third is the *inverse* case of Karpathy and nanoGPT: Karpathy is famous for too many things for a probe about him to surface nanoGPT specifically, while a probe about nanoGPT reliably surfaces him as the originator. Per-pair, per-model mention breakdowns are in Appendix D.

3.3.3 A worked case: Tianshou and its author

Jiayi Weng (author of the Tianshou reinforcement-learning library, then an OpenAI infrastructure engineer) is the sharpest single-entity illustration of the mechanism. Against eight OpenAI individual-contributor peers who share the affiliation but not a named artifact, he out-scores the peer mean by $+1.24\sigma$—and the lift is artifact-mediated: Tianshou's own NameRank exceeds his, and both survive translation to Chinese prompts unchanged (full comparison table in Appendix D).

A per-response signature points the same way: the panel models that mention Tianshou when asked about Jiayi Weng score roughly twice as high on him as those that do not (Appendix D)—but that comparison is observational, since the mentioning models are exactly the frontier-reasoning subset. The controlled version is the experiment below.

3.3.4 Context injection corroborates the mechanism

To manipulate the artifact mention directly, we re-probed all 11 pairs under a controlled intervention: configuration A uses a role-only disambiguating context for the creator; configuration B appends one clause naming the artifact (e.g., "creator of Tianshou"), holding everything else fixed. Figure 4b reports the per-pair lift (full table in Appendix D). Injection raises creator NameRank exactly where the observational analysis predicts: most for creators the panel barely stores, not at all for senior leaders whose recognition is already saturated; and by model tier the lift grows as capability decreases, so naming the artifact is load-bearing precisely for models that have not independently stored the creator.

One confound bounds the reading: because the injected clause names the artifact and the gold also names it, a model can raise coverage by echoing the injected token. The lift's concentration in low-baseline creators and the capability gradient both cut against pure echo (which would credit saturated leaders too), but the clean test—re-judging against an artifact-redacted gold—is registered for the next run. We therefore read the experiment as *consistent with* a causal named-artifact-amplification mechanism, with the measured lift as an upper bound (detail in Appendix D).

3.4 What Predicts NameRank

3.4.1 Among bibliometrics, h-index carries the signal—and still explains under half the variance

On the OpenAlex working-researcher cohort ($n = 771$, with bibliometrics from OpenAlex), the h-index explains roughly three times the NameRank variance that raw citations do, and citations add *zero* marginal predictive power once h-index is in the model (Figure 5). The dominance is strictly relative, not absolute: even the h-index fit leaves ∼60% of individual-level variance unexplained, shrinking the residual standard deviation only from 0.176 to 0.137. No bibliometric is a good individual-level predictor of recognition—consistent with the premise of Section 1 that the bibliometric residual is the quantity of interest. The pattern is stable out of sample under repeated cross-validation; full regressions are in Appendix E.

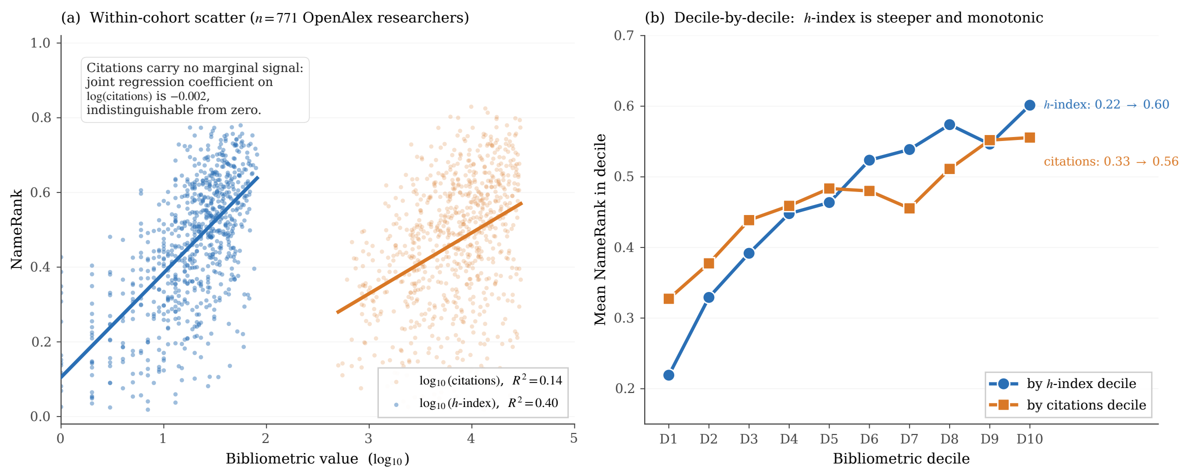

Figure 5. Among bibliometrics, h-index carries the signal ($n = 771$ OpenAlex researchers). **(a)** NameRank against $\log_{10}(h\text{-index})$ (blue) and $\log_{10}(\text{citations})$ (red): the h-index fit explains $R^2 = 0.40$ of within-cohort variance against 0.14 for citations, and the advantage persists out of sample (10-fold CV $R^2 = 0.38$ vs. 0.12; Appendix E). **(b)** Mean NameRank by decile of each bibliometric: the h-index ladder rises from 0.22 to 0.60 (near-monotonically, with a single dip at D9), while the citation ladder is much flatter. In joint regression, citations contribute zero marginal R^2. The claim is relative, not absolute: even the h-index fit leaves most individual-level variance unexplained (residual SD 0.137 against a total SD of 0.176)—the visible scatter in (a) is real, and that residual is the quantity NameRank is built to measure.

The mechanism is name-recurrence, not attribution weighting. h-index counts distinct works that each carry non-trivial citation mass—a sustained pattern of citable artifacts recurring under the author's name across the corpus. The natural alternative—that h-index wins by discounting per-paper name dilution in hyperauthored papers [Cronin, 2001]—fails: attribution-weighted citation measures barely improve on raw citations, while the raw count of distinct works tracks h-index closely (Appendix F). NameRank is a *name-recurrence* metric: it rewards how many distinct citable works repeat the author's name, not how cleanly attribution divides within any one paper.

3.4.2 Institution and country: the corpus-density gradient

Within the CS-faculty cohort ($n = 698$ across 320 institutions), affiliation explains ∼16% of within-cohort variance once the 320-category fit is penalized (adjusted R^2, ω^2, and intraclass correlation all ≈0.16; the naive categorical R^2 of 0.54 is a degrees-of-freedom artifact). What survives cleanly is the large-cell contrast: Stanford-affiliated faculty score about twice Tsinghua-affiliated (0.54 vs. 0.26; full institution table in Appendix G).

Aggregating by country (Figure 6; full table in Appendix G) shows the gradient at population scale. The firm, well-sampled contrast is the USA ($n = 302$) against China ($n = 30$) and India ($n = 10$), whose confidence intervals do not overlap the USA's; the apparent small-country leaders (Israel, Singapore, Sweden, at $n = 5$–7 each) sit above the USA in point estimate only and should be read as suggestive.

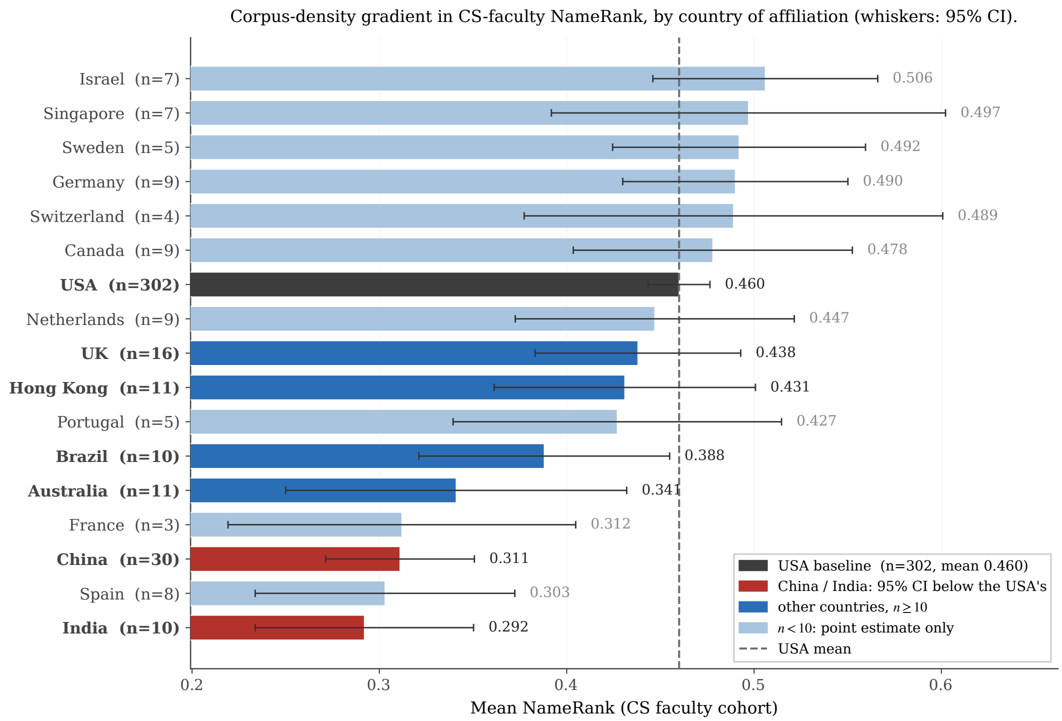

Figure 6. Country gradient in CS-faculty NameRank (whiskers: 95% CIs; faded bars: $n < 10$, point estimates only). The firmly-separated contrast is USA ($n = 302$, dark) vs. China ($n = 30$) and India ($n = 10$) (red), whose CIs do not overlap the USA's. The gradient tracks English-language web-text production per affiliation, not research quality: Tsinghua, Peking, and USTC carry global research reputations comparable to upper US institutions yet sit at the bottom.

The gradient is corpus density, not research quality. A young Tsinghua faculty member produces papers, lab pages, and talks as rigorous as a CMU counterpart's, but they propagate mainly through Chinese-language channels plus the English papers themselves, while the CMU counterpart's identical output additionally propagates through English social media, newsletters, podcasts, and a denser tutorial ecosystem. The CMU researcher's name-attribution mass per unit of research output is structurally higher; NameRank measures the result.

3.4.3 Prompt language routes retrieval

Two experiments separate what the *models* contribute from what the *prompt language* contributes. First, model origin matters less than refusal style: the Chinese-panel means run slightly above the Western-panel means across nearly every cohort, but the high-refusal and zero-refusal clusters each contain both Western and Chinese models—the dominant axis is a training-regime difference, not corpus coverage (released per-model statistics). Second, prompt language routes retrieval into language-specific corpus pockets: re-probing the 240-entity subset in Chinese lifts Chinese-named cohorts (DeepSeek-V3 authors +0.069) and suppresses English-coded entities, with Western and Chinese models shifting together (full cohort and per-entity tables in Appendix H). Our reading: even Western models hold Chinese-text pockets about Chinese-named entities, and the prompt language is the routing signal that unlocks them; inversely, an English-coded artifact is best recalled from English text, and a Chinese probe routes retrieval into thinner corpus regions.

Methodological implication. Within-entity scores are fairest in the entity's dominant-corpus language, and cross-entity comparisons must hold the prompting language fixed to control for the routing effect. All headline results in this paper use English prompts; the Chinese-prompt scores are released as a separate dataset.

3.5 News Events: The Attention Ledger

Every predictor so far proxies corpus exposure after the fact—bibliometrics for researchers, affiliation for faculty. News events invert the measurement problem: for an event, the attention the world paid is itself a public, time-stamped record. We therefore built a separate calibration cohort of 258 discrete news events from 2021–2023, harvested from the English Wikipedia year and country-year pages, machine-classified for date, region, and category, and stratified to span four orders of magnitude of recorded attention across eight world regions (construction in Appendix I). The window keeps the events inside the panel's training cutoffs, so recognition failures cannot be corpus-timing artifacts (cf. Appendix M); the three oldest models' cutoffs trail only the window's final weeks, and excluding them—or the 22 late-2023 events—moves no estimate (Appendix I). Each event is probed and judged by the unchanged pipeline of Section 2: gold answers are the event article's intro, contexts are deterministic type/place/date phrases (*"a natural disaster that occurred in Turkey and Syria in February 2023"*).[2]

The attention ledger is the article's daily en-Wikipedia pageview series over the 365 days from event onset [Mestyán et al., 2013]: *total attention* (summed views), *peak attention* (maximum daily views), and *effective duration* (total/peak, in days—the length of a rectangular pulse with the same peak and total). Pageviews are preferred to Google Trends because they are absolute, archived, and reproducible rather than normalized and rate-limited; a GDELT news-volume cross-check [Leetaru and Schrodt, 2013] is described in Appendix I. Table 2 shows the cohort's texture; Figure 7 the two central results.

Recognition tracks the ledger, through accuracy. NameRank rises log-linearly in total attention and refusals fall from 12% to 1% across the attention range (Figure 7a; $R^2 = 0.10$ in-sample and under cross-validation alike, 0.17 after controlling gold density, which suppresses the slope because bigger events have denser Wikipedia intros). The channel is accuracy more than coverage (Appendix I): marginal attention does not make models recite more of an event—it makes what they say *correct*, converting fluent confusion into specifics. This is the one cohort where the exposure side of the corpus-propagation account is directly measurable, and the account holds where it can be checked.

[2] Nine main-run models could no longer be run at event time (2026-07 provider churn plus one exhausted vendor account); the eight with released replacements were substituted by the representative newly released models evaluated in the IKP revision [Li, 2026], giving a 36-model panel (28 survivors + 8 replacements). Neither change matters: main-run NameRank recomputed on the surviving sub-panel correlates $r = 0.993$ with the released scores, and the replacement models rank events at $r = 0.905$ with the survivors (Appendix I).

Table 2. Selected events across the attention range. *Views* is total first-year en-Wikipedia pageviews of the event's article; *peak* the maximum daily views; *dur.* the effective duration total/peak. † marks recurring-series instances (an article with the same base name exists under another year; Section 3.5, name uniqueness). Full cohort in the release.

Event	Views (yr 1)	Peak/day	Dur. (d)	NameRank
Siege of Mariupol	2.6M	58K	46	0.708
2021 Suez Canal obstruction	1.2M	150K	8	0.664
Assassination of Shinzo Abe	2.2M	558K	4	0.595
2022 FIFA World Cup†	39.3M	1.9M	21	0.471
2022 Russian invasion of Ukraine	52.0M	2.1M	25	0.449
Hurricane Ian	1.8M	121K	15	0.361
2022 Malaysian general election†	1.3M	271K	5	0.174
2021 Guangzhou bombing	12.8K	3.2K	4	0.092

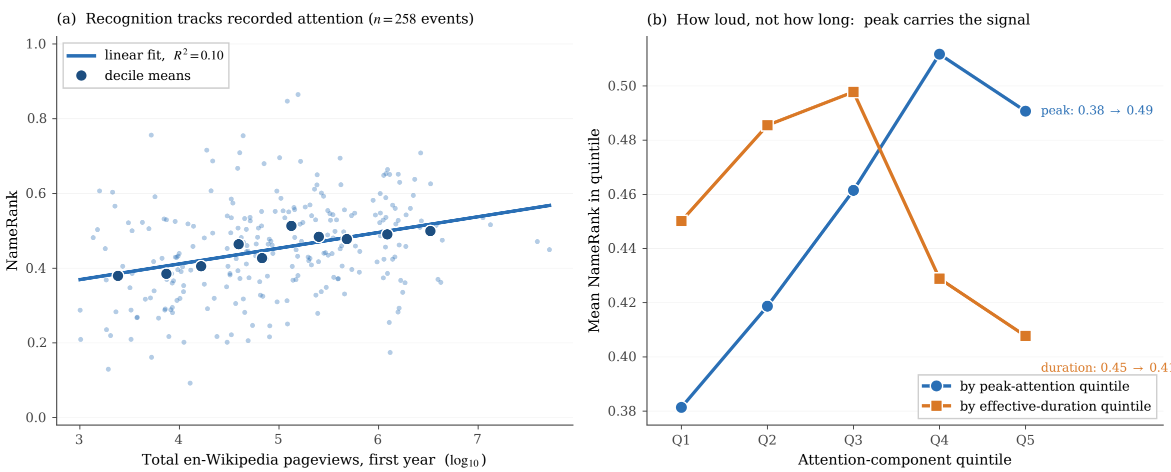

Figure 7. The news-event calibration cohort ($n = 258$, 2021–2023). **(a)** NameRank against $\log_{10}$ total first-year en-Wikipedia pageviews; small dots are events, large markers decile means ($R^2 = 0.10$; 0.17 with gold-length control). **(b)** Quintile ladders for the two log-components of total attention: peak attention rises 0.38 → 0.49; effective duration is flat-to-negative (0.45 → 0.41). In joint regression the standardized coefficient on duration is -0.008 (se 0.009) against $+0.042$ (se 0.008) on peak.

How loud, not how long. Total attention factors exactly into peak × duration, so its log decomposes into two channels—and only one carries signal (Figure 7b): the standardized coefficient on log peak is $+0.042$ (se 0.008) against -0.008 (se 0.009) on log duration, an asymmetry that survives recurring-name and gold-length controls (Appendix I). We had expected the opposite—the temporal analogue of the h-index result—and the failure is informative. A story's peak marks how widely its name was *reproduced* at climax: wire copy, front pages, explainers, each a distinct indexable document minted in days; whether attention then decayed in a week or smoldered for months changes how long people kept *reading*, but mints little additional name-bearing text. Recognition follows document production, not audience retention: h-index wins for researchers and peak wins for events for the same reason—name-recurrence across distinct documents, measured against different clocks.

Templated names dilute. 106 of the 258 events are *recurring-series instances*—strip the year and the same base name exists under another year (elections, tournaments; flagged mechanically, Appendix I)—and at matched attention they score -0.061 (se 0.015) below singular events. The 2022 Malaysian general election drew 1.3M first-year views and scores 0.174: the models know *Malaysian general elections* exist but cannot bind facts to the 2022 edition, exactly the name-collision failure that IKP's name-uniqueness mechanism predicts [Li, 2026], while a one-off name (*Owo church attack*, one tenth the attention) scores 0.847.

What does not move the score. At matched English-Wikipedia attention, region effects are small—consistency, not contradiction, with Section 3.4.2: for events the corpus-density gradient operates through *how much English attention the ledger records in the first place*, and conditioning on the ledger absorbs it. Panel composition does not move it either: the eight 2026-vintage replacement models rank the events at $r = 0.905$ with the survivors—the entity-not-panel property replicated on entirely new models—and event year shows only a small vintage echo in the direction of the drift documented in Appendix M. Finally, attention has a ceiling: the cohort averages in the CS-faculty band, and even the 2022 Russian invasion of Ukraine, at 52M first-year views, scores 0.449, far below the universal zone. Event golds are dense with one-off specifics that no 150-word response can cover; the ceiling is a gold-density property (Appendix O), but the ordering underneath it is what the ledger predicts (full regressions and region table in Appendix I).

3.6 Boundary Condition: Reach Without Name Propagation

One entity in the diagnostic set falsifies the tempting reading of NameRank as "impact." tuixue.online is a Chinese-language website that monitored US F1-visa appointment availability between 2019 and 2024, with a peak user base near one million applicants who relied on it during the post-COVID visa backlog. By any conventional reach metric—active users, downstream effect on visa appointments—it exceeds nanoGPT, whose audience is tens of thousands of developers. Its NameRank is nonetheless a third of nanoGPT's (Table 3), and the dissociation is not an English-corpus artifact: the Chinese-prompt score is essentially identical, so the name-attribution absence is symmetric across both training corpora.

Table 3. Reach and recognition dissociate. tuixue.online has the largest human reach in this comparison set and the lowest NameRank; its Chinese-prompt score (0.262) matches its English-prompt score, so the gap is not a language artifact.

Entity	NameRank	Approx. human reach
nanoGPT	0.707	∼10K developers
FlashAttention	0.607	academic + library users
ReAct (paper)	0.578	academic
vLLM	0.527	∼100K developers
LangChain	0.502	∼100K developers
tuixue.online	**0.250**	**∼1M visa applicants**

The mechanism is attribution convention. tuixue.online circulated in chat groups and forums referenced functionally ("the site has visa slots open") with no creator name attached in indexable form, while open-source projects of comparable utility carry their creators' names on READMEs, talks, paper bylines, and tutorials across thousands of indexable documents. This is the boundary condition of the construct: NameRank measures *indexed corpus presence of a name*, not human utility. The two correlate wherever creators attach names to work, and dissociate for utility-driven artifacts that spread by word of mouth under anonymous-by-convention attribution. The dissociation is not a flaw to be patched; it is a demonstration that the metric measures something specific and falsifiable.

3.7 Self-Reported Recognition: Asking a Model Is Not Measuring It

An obvious shortcut would collapse the instrument to a single prompt: instead of probing and verifying, ask a frontier model which names it recognizes. We test whether the shortcut works—and, underneath it, whether models have introspective access to the quantity NameRank measures behaviorally. Three panel models (GPT-5.5, Claude Opus 4.6, Gemini 3.1 Pro, at main-run settings) each judged the same 3,350 name pairs over 430 entities (400 stratified across the NameRank range plus the 30 fictional controls of Appendix O), stating which of the two they know more about, that they know both equally, or that they recognize neither; a tie-extended Bradley–Terry model [Bradley and Terry, 1952, Davidson, 1970] converts each model's verdicts into a self-reported ranking. Design, estimation, and full tables are in Appendix J.

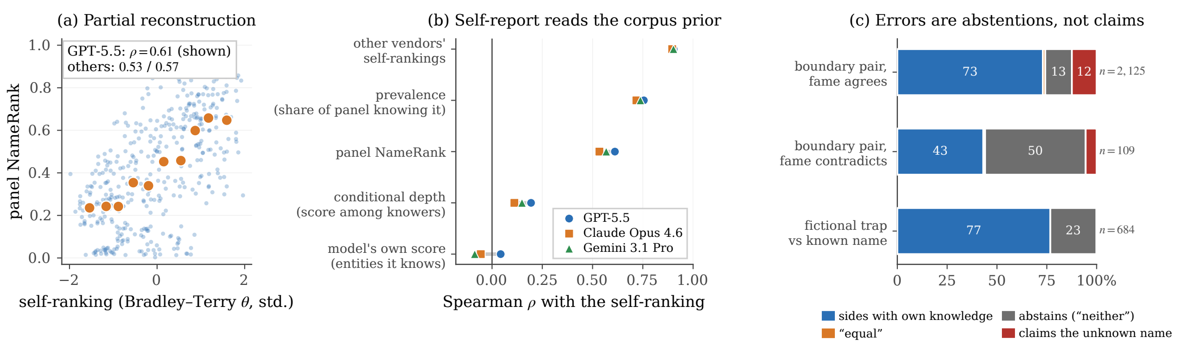

Figure 8. Self-reported recognition (three frontier panel models, 3,350 shared pairs). **(a)** Bradley–Terry self-ranking against panel NameRank (GPT-5.5 shown; orange markers are decile means): $\rho = 0.53$–0.61 across the three models. **(b)** What the self-ranking tracks: agreement with the *other vendors'* self-rankings ($\rho \approx 0.90$) and with the prevalence component of NameRank—the share of the 37-model panel that knows the entity ($\rho = 0.72$–0.76)—is far higher than with the reporting model's *own* behavioral scores among entities it demonstrably knows ($|\rho| \leq 0.09$). **(c)** Verdict shares (pooled over models) on boundary pairs whose one side the model behaviorally knows, split by whether corpus fame agrees with the model's own knowledge, and on fictional-trap pairs: when fame contradicts its own knowledge the model abstains half the time, and the fictional side is claimed in 0.1% of trap pairs.

The shortcut half-works, and the failing half is diagnostic (Figure 8). Self-reported rankings recover the panel ordering only partially—a usable prior, not a substitute for measurement. More telling is *what* self-report reads (panel b): each model's self-ranking tracks how *widely* a name is known across the panel, is essentially uncorrelated with the model's own behavioral record among entities it demonstrably knows, and agrees with the other two vendors' self-rankings far more closely than with the model's own scores. When its own knowledge contradicts corpus fame (panel c), the model mostly abstains rather than reading its own state. Introspection, in short, returns a shared corpus prior, not a reading of the model's own weights [Kadavath et al., 2022, Yin et al., 2023].

In one direction, however, the claims are precise, echoing the silent-rather-than-misleading property of Section 2.9: opposite a name the model demonstrably knows, the fictional side of a trap pair was preferred exactly once in 684 presentations, and fictional-vs-fictional pairs drew "neither" almost without exception. Self-reported recognition errs by under-claiming, not fabrication: trustworthy when asserted, insufficient as a measurement.

4 Robustness and Validation

The findings above rest on three design choices—an LLM judge over embedding similarity, a multiplicative coverage×accuracy rule, and open-ended over closed-factoid probes—and on the headline numbers not being artifacts of wording, model vintage, judge family, or gold-answer construction. Table 4 summarizes the validation battery; Appendices K–P give the full evidence. Three results carry most of the weight.

Table 4. The validation battery: each threat to the findings, the check run against it, and the outcome. Full tables and figures in the referenced appendices.

Threat	Check	Outcome	
Score reflects panel composition, not the entity	Variance decomposition over all 211,603 records; disjoint-panel re-score of the event cohort	Entity variance ≈3.3× model variance; eight 2026 replacement models rank events at $r = 0.905$ with the surviving panel (§3.5)	App. K
Fluent hallucinators inflate scores	Refusal and generosity structure across the panel	Accuracy axis zeroes wrong-person bios; refusal rate tracks the silent zone	App. K
A judge-free embedding score would suffice	Judge-vs-embedding comparison, 175K records	Embedding similarity is flat across judge scores and credits hallucinations	App. K
Findings are an artifact of probe wording	Four-template re-probe, 29,600 records	Rankings invariant ($r \geq 0.93$); absolute levels are template-conditional	App. L
The silent zone is corpus timing, not obscurity	Training-cutoff natural experiment	Matched difference-in-differences ≈0; silence is intrinsic	App. M
Findings depend on the judge's model family	Three-judge re-grade, 615 records	Judges agree ($r = 0.87$–0.93); small own-family lift moves no finding	App. N
Unknown names could score above zero	30 fictional entities through the full pipeline	Floor ≈0.04 (dense golds), ≈0.20 (boilerplate golds); adjusting for it *widens* the credential gap	App. O
Gold length, context leakage, Wikipedia presence, or web attention drive the results	Dedicated ablations and regressions; 24-month pageview baseline	All four rejected	App. P

The scoring rule is load-bearing. Across all records, the entity factor explains over three times the variance the model factor does, and an entirely disjoint replacement panel reproduces the event ordering—the score measures the entity, not the fleet. The refusal structure shows why the multiplicative rule matters: the zero-refusal models are fluent hallucinators whose confident wrong-person bios earn high coverage but near-zero accuracy, so their product contributes nothing, while embedding similarity—flat across judge-score buckets—would credit exactly those bios as recognized (Appendix K).

Orderings survive wording, vintage, and judge; levels do not. Re-probing under three paraphrased templates leaves the entity ordering and every cohort-level finding intact while shifting absolute levels, so NameRank values are template-conditional (Appendix L). A training-cutoff natural experiment on two contemporaneously-published author cohorts shows the silent zone is intrinsic rather than corpus timing—but also reveals a $\sim$0.13/yr cross-vintage drift, so longitudinal claims must be made on within-run *relative* gaps (Appendix M); the thresholded behavior itself is what the transformer fact-storage literature predicts [Geva et al., 2021, Petroni et al., 2019, Kandpal et al., 2023, Zhang et al., 2025]. And a three-judge re-grade (Gemini, Claude, GPT-5) preserves every finding, with only a small own-family scoring lift that shifts per-vendor tables (Appendix N).

Floors are calibrated and confounds rejected. Thirty fictional entities establish the noise floor: near-zero for dense golds, elevated for boilerplate golds—and floor-adjusting *widens* the credential gap (Appendix O). Dedicated ablations reject gold-answer length, context leakage, and Wikipedia presence or attention as alternative explanations: NameRank is not a has-a-Wikipedia-page flag—the binary flag explains $\sim$2% of long-tail variance against $\sim$40% for h-index—and it is not pageview rank, which is undefined for 71% of entities and explains $R^2 = 0.06$ where defined (Appendix P).

5 Discussion

5.1 What We Have Built and What We Have Not

NameRank is a continuous cross-domain instrument with empirical threshold cleanliness, validated external correlates (h-index, institution, the event attention ledger), and mechanism-level structure. It is suitable for cross-domain placement on a common scale, for tracking name propagation into training corpora over time, and for measuring the recognition channel that bibliometrics miss. It is not a quality metric (it measures what models absorbed, not what is worth absorbing), not an attribution system (the name, not the deed), and not predictive of anything the corpus has not yet recorded.

5.2 The Credential Treadmill Thesis in Context

The credential treadmill does not contradict the olympiad-outcome literature, which finds medals causally raise subsequent *academic productivity* [Agarwal and Gaule, 2020, Lubinski et al., 2020]; our dependent variable is different. When it is name-propagation into LLM corpora rather than paper count, the credential per se is a near-zero predictor—continued production of named, indexable artifacts is what propagates. The credential still signals ability and motivation; only its role as a name-propagation channel has shrunk. In the language of signaling theory [Spence, 1973], the LLM-corpus channel transmits not the credential itself but the named, indexable production that may follow it.

5.3 What Clock Does NameRank Read? Lifetime vs. Current-Term Recognition

A recognition score could measure two different quantities: *cumulative* presence—everything a name has deposited in the corpus over a career—or *current-term* salience, what the world is paying attention to now. Three findings established independently above jointly place

NameRank in the first category. Within IMO gold medalists, medal year is uncorrelated with the score (Pearson -0.04 over 2005–2015; Appendix B): recognition neither fades for older medalists nor spikes for recent ones. MSRA fellowship cohorts show the complementary shape—five-year-cohort means rise from 0.312 (2005–2009) to 0.382 (2015–2019) and dip only for the 2020–2024 fellows—exactly what an integral over post-credential production looks like when the newest cohorts have not yet had time to produce (Appendix B). And on news events, where attention is directly recorded, the score loads on *peak* salience rather than persistence (Section 3.5): what matters is how many name-bearing documents were minted, not how long the audience stayed. Read together: NameRank is a cumulative ledger of name-bearing text, snapshotted through the current model fleet—a lifetime measure weighted by documentation density, not a barometer of current attention. Current-term change is accessible only longitudinally, as within-run relative gaps across successive fleets (Section 5.8).

5.4 The Artifact > Creator Inversion and the Anonymity-of-Tooling Problem

For independent creators of well-named single artifacts the tool out-ranks its maker (Section 3.3), and the context-injection experiment is consistent with this being a causal effect (modulo a coverage-echo confound we bound but do not fully remove, Section 3.3.4). The mechanism is attribution-direction asymmetry: creator-bios feature their creations, but artifact-discussion text (OSS docs, tutorials, papers) names the artifact far more often than the creator. The actionable consequence is uncomfortable but concrete: for propagating a name through the LLM channel, building a clearly-named tool beats writing many papers—yet the lift accrues to the tool's name first and to the creator only through it, so a tool buried under generic corporate branding yields no personal NameRank.

5.5 The Corpus-Density Gradient and Its Limits

Section 3.4.2 documents a twofold Stanford–Tsinghua recognition gap that recurs at country scale, with the well-sampled USA-over-China-over-India ordering as its firm backbone and the small-country leaders as suggestive only. The mechanism is English-corpus density; the consequence is structural inequality in name propagation per unit of research output.

Three caveats: the bottom-of-ladder institution samples are small ($n = 4$); uncorrected researcher mobility inflates the US average and depresses China's; and the gradient is downstream of the indexing and crawling decisions of training-data pipelines, so it could in principle be reduced by changing them. Within the current corpus regime, it is a substantial structural effect.

5.6 Inherited Demographic Biases

NameRank inherits the demographic biases of its judge and probed models, and measures them faithfully rather than correcting them. Li [2026] documented a $\sim 2\times$ recognition attenuation for common East-Asian surnames; we find a second, smaller effect on first-name-inferred gender: within the researcher cohorts—where institution matching and h-index-decile adjustment are available and survive—man-coded names outscore woman-coded names by roughly 0.03–0.05 (Appendix Q). We restrict the claim to that controlled comparison: the much larger raw gaps in cross-profession cohorts reflect real fame differences among the sampled individuals, not a NameRank artifact. As with the surname effect, the gender gap is an inherited property of the corpus-and-judge stack that NameRank reports, not a calibration target; downstream uses must account for it.

5.7 Limitations

Appendix R tabulates every caveat with its direction and mitigation. The material ones: absolute levels are wording- and gold-conditional while rankings are robust; the full run uses a single judge with a small own-family lift; only within-run *relative* gaps are comparable across model fleets ($\sim$0.13/yr vintage drift); and the context-injection lift is an upper bound on named-artifact amplification.

5.8 Registered Predictions for Future Re-Runs

NameRank is a snapshot, so a re-run each release cycle yields a longitudinal trajectory. Because of the $\sim$0.13/yr vintage drift, we state predictions as within-run *relative* gaps (so the drift cancels) and register four falsifiable ones for the 2027 fleet: (i) the IMO-gold-vs-OpenAlex gap widens from -0.102 to ≤ -0.13 within 12 months (named-artifact-attached careers accumulate text, credential-only cohorts do not); (ii) the median artifact-minus-creator delta on independent-creator pairs grows from $+0.09$ to $\geq +0.14$ within 18 months; (iii) the Stanford-vs-Tsinghua gap narrows by $\geq$ 25% but does not close within 24 months as Chinese-language training data enters frontier models; (iv) on a re-run of the news-event cohort (Section 3.5)—the cleanest of the four, since its exposure side is frozen in the pageview archive—the 2023-vs-2021 vintage gap (-0.049 at matched attention) shrinks by at least half within 18 months as post-event derivative text accumulates.

6 Related Work

6.1 Relation to the Predecessor Instrument

The direct predecessor of this work is the recognition analysis in the IKP benchmark [Li, 2026, §5.7], which established the findings this paper builds on: on 345 CS-researcher probes, log(citations) explains only $\sim$35% of recognition variance, with the remainder attributed in a controlled audit to *named-artifact amplification*, *name uniqueness* (common East Asian surnames attenuate recognition by $\sim 2\times$), and *subfield-ecosystem density*; and on 557 Wikidata-grounded probes, domain-specific recognition multipliers trace to the structure of web discourse around each fact type rather than to entity prominence. IKP nonetheless treated recognition as a benchmark byproduct. NameRank differs in each dimension that matters for an impact metric: a continuous $[0, 1]$ score over a 37-model panel (vs. a six-landmark tier ladder calibrated for parameter estimation); 5,719 entities across 54 cohorts spanning industry, OSS, and mid-tier figures (vs. 345 researchers and 557 Wikidata probes); open-ended probes scored on multiplicative coverage$\times$accuracy (vs. closed-factoid binary verdicts); h-index rather than raw citations as the validated bibliometric correlate; and cross-model variance retained as signal rather than absorbed into a tier label.

6.2 Scientometric Measures of Researcher Impact

The classical instruments for researcher impact—citation count, h-index [Hirsch, 2005], journal impact factor—measure intellectual reach within the academic publication system, and are increasingly understood as failing under pressures relevant here: authorship-list inflation and uneven attribution [Larivière et al., 2019], the *hyperauthorship* problem of pricing contributions to thousand-author papers [Cronin, 2001] (a GPT-4 co-author's citation count is mechanically large but says nothing about name-recognition), prestige-hierarchy biases spanning an order of magnitude across institutional tiers [Wapman et al., 2022], and within-researcher variance that motivates decompositions like the Q parameter [Sinatra et al., 2016].

None of these instruments cross the academic boundary cleanly. Citation count does not measure Sam Altman, Aravind Srinivas, or a Walmart CTO; h-index does not measure the creator of nanoGPT. Bespoke domain metrics (Google PageRank for websites, GitHub stars for OSS, Spotify monthly listeners for musicians, IMDb scores for film) do not interconvert. Collective-attention metrics do—search volume, Google Trends, and Wikipedia pageviews place any entity on one axis, and pageviews in particular are an established salience proxy [Mestyán et al., 2013, Ripberger, 2011]—but they count queries and article traffic rather than verified knowledge, and they are empirically near-orthogonal to NameRank: undefined off Wikipedia, weakly predictive where defined, and negatively correlated within technical cohorts (Appendix P). We use attention records as calibration input on the event cohort (Section 3.5), not as the measurand: NameRank reads the *outcome* of the training pipeline—what deployed models can correctly state—rather than the exposure that feeds it. The cross-domain measurement of eminence has a long prior tradition in *historiometry* [Simonton, 1999], which quantifies the reputational standing of eminent individuals from archival traces; NameRank can be read as a historiometric instrument whose archive is the frontier-model training corpus rather than biographical dictionaries and citation indices. For impact assessment that spans academia, industry, OSS, and the long tail of solo operators, a cross-domain instrument is needed; NameRank is the first we are aware of that reads this archive directly.

6.3 LLM-as-Judge for Quality and Impact

Thelwall [2024, 2025b,a] use ChatGPT to score paper *quality* against the UK REF rubric; Kousha and Thelwall [2025] extend to societal influence; Liu et al. [2026] forecast citation counts. These differ from NameRank in two respects: they ask the LLM to *judge* quality, where we ask what the model has *absorbed*; and they average within a single model across iterations, where we average across 37 models, treating cross-model variance as signal.

A separate line asks whether models *know what they know*: self-assessed confidence is well calibrated on aggregate question answering [Kadavath et al., 2022] but degrades at the boundary of a model's own competence [Yin et al., 2023], and entity popularity modulates factual reliability [Mallen et al., 2023]. Section 3.7 adds an entity-level counterpart: pairwise self-reported recognition, aggregated with the Bradley–Terry machinery familiar from Chatbot Arena [Chiang et al., 2024], tracks the corpus salience shared across vendors rather than the reporting model's own verified knowledge.

The known LLM bias most relevant to our setting is the *Matthew effect* [Merton, 1968] as it propagates through LLM training corpora: frontier models recall already-highly-cited work disproportionately, inflating the rich-get-richer dynamic familiar from the bibliometrics literature. We inherit this bias and treat it as a known property of the measurand—NameRank measures *what frontier models have absorbed*, which is itself shaped by the Matthew dynamic. The bias is part of what we measure, not a flaw to be corrected.

6.4 Credentials and Long-Run Outcomes

A longitudinal literature finds that early mathematical talent and olympiad medals predict—and causally raise—later *academic* productivity: the 50-year SMPY study [Lubinski et al., 2020], a medal-cutoff natural experiment [Agarwal and Gaule, 2020], and work documenting substantial within-elite heterogeneity [Güllich et al., 2025]. Our credential-treadmill finding does not contradict this literature; it changes the dependent variable from academic production to LLM-mediated name recognition and finds that the credential per se carries little of it (Sections 3.2, 5.2).

7 Conclusion

NameRank operationalizes the 65% recognition-variance residual that Li [2026] characterized but did not measure: a continuous, cross-domain, hallucination-resistant recognition score over a 37-model panel. Its findings each subvert a default expectation—olympiad credentials no longer propagate names, named tools out-rank their makers, the structure of citation production beats its volume, on news events, where world attention is directly recorded, peak salience rather than persistence drives propagation, and asking a model to rank its own recognition returns the shared corpus prior rather than its own knowledge state—and each is mechanistically grounded and survives a battery of robustness checks. The broader claim is that human curiosity about people and artifacts is now re-mediated by a few frontier models, and whether a name has propagated into their training corpora is itself a measurable quantity. Three uses follow: a cross-domain impact metric (a Walmart CTO at 0.52 is directly comparable to a Stanford professor at 0.54), a longitudinal tracker of how discourse reaches training pipelines, and a normative indicator for career strategy (build and name your own tool). All probes, gold answers, responses, and code are released at `https://github.com/19PINE-AI/namerank`; companion site `https://01.me/research/namerank`.

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The appendices are organized in three blocks. Appendix A completes the instrument specification of Section 2. Appendices B–J give the full tables and case detail behind each finding of Section 3, in the order the findings appear. Appendices K–Q give the evidence behind the validation battery of Section 4, one appendix per threat, followed by the limitations summary (Appendix R).

A Probe, Panel, and Cohort Specification

Why disambiguation does not contaminate scoring. The probe context names role, affiliation, and field—never specific contributions, papers, dates, or named artifacts. A model that recognizes the entity produces specific facts, which the judge scores against the gold; a model that does not produces generic statements about the role, which the judge assigns low coverage because the rubric credits only *specific* named facts (Section 2.4). The sharpest empirical check comes from the GPT-5 system-card author cohort, whose disambiguating context is identical across all 80 entities: 71% of probes nonetheless return "unknown" rather than a generic in-context biography. A dedicated context-ablation experiment (Appendix P) confirms the same conclusion.

Table 5. The 37-model panel, by vendor class. Thinking-mode variants are used wherever available.

Western (n=23)	Chinese (n=14)
GPT-5.3, GPT-5.4, GPT-5.5-think, GPT-OSS-20b-think	DeepSeek-V3.2-think, V4-flash-think, V4-pro-think
Claude Opus 4.6-think, Sonnet 4.6-think	Qwen3-8b-think, 32b-think, 235b-a22b-think, 3.5-397b-a17b-think
Gemini 2.5-pro-think, 3-flash-think, 3.1-pro	GLM-4-32b, 4.7-think, 5.1-think
Llama-3.1-8b, 3.2-1b, 3.3-70b, 4-Maverick	Kimi-K2, K2.6-think
Mistral-Large, Medium 3.1, Small 24b, Ministral 3b	Ernie-4.5-300b-a47b
Gemma-3-4b, 3-12b, 4-31b	MiniMax-M2.7-think
Grok-4, Grok-4.20-think, Phi-4	

Table 6. Entity cohorts by top-level group. Full per-cohort breakdown in the released cohort specification.

Group	*n*	Representative cohorts
Credentialed individuals	801	IMO/IOI/CMO/NOI/CPhO medalists, Putnam, ICPC, Rhodes, MSRA Fellowship
Working academics	1,628	CS faculty; OpenAlex researchers with 500–30,000 citations
Industry figures	248	founders, VCs, YC companies, AI-policy officials
Named artifacts	1,332	foundation models, OSS projects, papers, benchmarks, named methods
Mid-tier cross-profession	1,461	writers, athletes, actors, chefs, journalists, podcasts, books
Diagnostic reference set	37	hand-curated public figures, engineers, and artifacts
Special cohorts	212	GPT-5 system-card authors, DeepSeek-V3 authors, conferences, awards

B Credential Ladder Detail

Table 7 gives the full credential ladder summarized in Figure 3.

Table 7. Credential ladder. The OpenAlex long-tail researcher cohort ($n = 771$, citations 500–30,000) supplies the working-researcher baseline. The two rows marked † are large-team co-authorship cohorts shown for context, not credentials, and are excluded from the "7 of 9" count.

Credential	n	Mean NameRank	vs. baseline
ICPC World Finalist gold	18	0.610	+0.146
Putnam top-25 fellow	35	0.608	+0.144
OpenAlex long-tail researchers (baseline)	**771**	**0.464**	—
NOI China gold	29	0.368	−0.096
IMO gold (2005–2015)	**197**	**0.362**	**−0.102**
MSRA PhD Fellowship	237	0.343	−0.121
IOI gold	68	0.341	−0.123
Rhodes Scholarship	36	0.315	−0.149
CMO China gold	131	0.302	−0.162
CPhO China first prize	50	0.182	−0.282
DeepSeek-V3 paper author†	69	0.178	−0.286
GPT-5 system-card author†	80	0.042	−0.422

Career-track split. Within IMO gold medalists, US-tracked academic/industry careers—the subset most comparable to the citation-floored OpenAlex baseline—score ∼0.50; non-Anglophone-tracked golds score ∼0.30. The medal-and-name pair propagates only where the post-medal career produces English text.

Temporal patterns. Within MSRA fellows, five-year-cohort means rise from 0.312 (2005–2009) to 0.337 (2010–2014) to 0.382 (2015–2019) and dip to 0.343 (2020–2024), consistent with recent fellows having had less time for post-fellowship production to propagate. Within IMO 2005–2015, Pearson(year, NameRank) $= -0.042$, indistinguishable from zero: the credential signal neither strengthens nor decays within the ten-year window.

C Medal Tiers: The Complete NOI 2009 Roster

The credential ladder above measures *gold* medalists only, so it cannot say whether the medal's grade carries signal. To test this we probed the complete medal roster of a single contest — NOI 2009: every gold, silver, and bronze medalist (103 in all) — with the pipeline otherwise unchanged. The roster comes from the official CCF medal list, cross-validated line-by-line against the community OIerDb database (identical names, schools, and scores); the released files include each medalist's contest score and rank. The twenty golds re-enter with their main-run probes and gold answers verbatim; the new silver and bronze entities receive gold answers built from the same boilerplate recipe with only the tier band changed, so gold length and density are matched across tiers and the short-boilerplate synthetic-null floor (Appendix O) applies equally to all three. Probe names follow the main run's romanization conventions and carry the Chinese characters, which — together with the year-and-school context — disambiguate medalists whose Pinyin collides with common names. The run uses the 36-model 2026 panel of the news-event cohort (Appendix I); the 63 entities shared with the released main run reproduce their released ordering almost exactly (Spearman $\rho = 0.915$, mean shift -0.006).

The ladder is a cliff. Gold and non-gold are separated by a large, unambiguous gap, while silver and bronze are statistically indistinguishable (Table 8). Every gold medalist clears

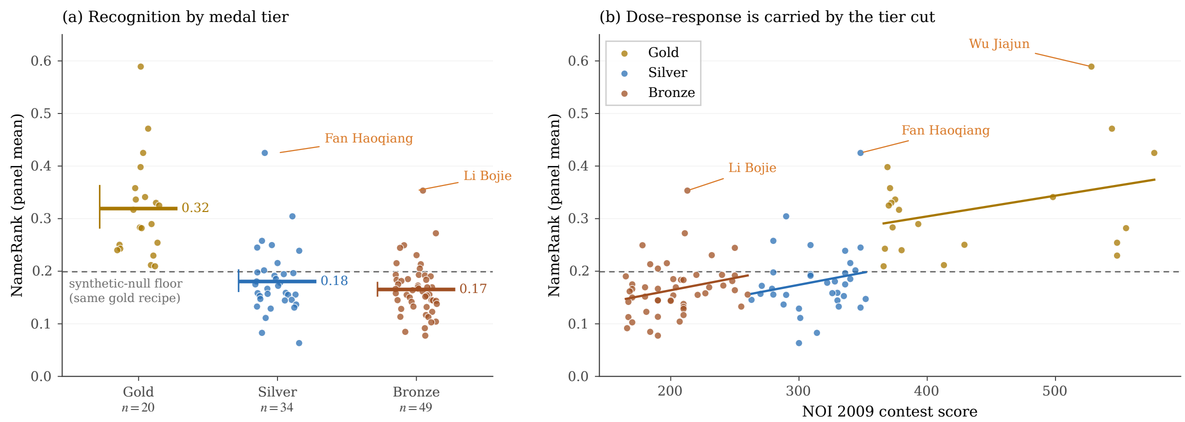

Figure 9. The complete NOI 2009 medal roster. **(a)** Per-medalist NameRank by tier (dots), with tier means and bootstrap 95% CIs, against the synthetic-null floor for this gold recipe. **(b)** NameRank against the contest score that determined the tiers, with within-tier least-squares fits. Callouts mark the medalists discussed in the text.

Table 8. NOI 2009 medal tiers. "Above floor" counts medalists exceeding the 0.199 short-boilerplate synthetic-null floor. Kruskal–Wallis across tiers $H = 39.3$ ($p < 10^{-8}$); gold vs. silver Cliff's $\delta = +0.85$ ($p < 10^{-6}$); silver vs. bronze $p = 0.25$. Pooled NameRank-on-contest-score $R^2 = 0.40$; within tiers it collapses to 0.11 (gold), 0.04 (silver), 0.07 (bronze).

Tier	n	Mean [95% CI]	Median	Above floor
Gold	20	0.319 [0.281, 0.362]	0.303	20/20
Silver	34	0.180 [0.160, 0.204]	0.169	8/34
Bronze	49	0.165 [0.152, 0.179]	0.162	8/49

the synthetic-null floor; the silver and bronze cohort *means* sit at or below it. Read against Property 2 of the metric, the silver and bronze rosters are not weakly recognized — as cohorts they are silent, with an individually recognized tail. The mechanism matches the corpus-density account of Section 3.2: the ∼20-person national gold roster is the documented one — admissions coverage, competitive-programming lore, team-selection reporting — while silver and bronze rosters barely mint name-bearing English text at all.

The cut matters; marginal points do not. Because tier is a threshold function of contest score, a pooled regression of NameRank on score looks respectable, but nearly all of it is the gold/non-gold cut: within tiers the explained variance collapses (Table 8, Figure 9b). Recognition does not accrue per point; it accrues on crossing the line into the documented roster.

The recognized tail is the treadmill again. The silver and bronze medalists who do clear the floor are exactly the ones whose *post-medal* careers produce named artifacts: the top-scoring silver medalist beats the gold-tier *mean* on the strength of a prominent computer-vision research career (Fan Haoqiang), and the top bronze medalist ranks near the bottom of the contest's score column yet seventh of 103 in recognition (Li Bojie, a systems researcher and entrepreneur — and the author of this paper, included with the same boilerplate gold as every other medalist). Within tiers, the medal-year cohort's gender gap also disappears (0.173 vs. 0.164 for non-gold men and women; the 2009 gold roster has no women), consistent with the controlled comparison of Appendix Q.

Levels are recipe-conditional. The same person appears in this run under three gold recipes — the medalist boilerplate (0.35), a fellowship-cohort boilerplate (0.41), and the dense diagnostic-reference biography (0.12 on this panel; 0.09 in the released run, the value shown in Table 1) — and spans a threefold range across them, despite identical model knowledge: a model that recognizes the person covers the short medalist gold completely, while even a detailed correct response covers only part of the 150-word biography. This is the Appendix O caution in within-person form: NameRank levels are comparable only within a gold recipe, which is why the tier ladder above holds the recipe fixed.

D Artifact–Creator Pairs: Inversion, Injection, and Asymmetry Detail

This appendix gives the full tables behind Figure 4 and the Tianshou case study (Section 3.3).

D.1 The inversion and injection tables

Table 9. The artifact-over-creator inversion. NR_c and NR_a are creator and artifact NameRank; $C \to A$ is the fraction of responses about the creator that mention the artifact, $A \to C$ the reverse. Boldface: artifact exceeds creator.

Creator	NR_c	Artifact	NR_a	$\Delta(a-c)$	$C \to A$	$A \to C$
Simon Willison	0.594	**Datasette**	**0.899**	**+0.305**	0.54	0.89
Dario Amodei	0.584	**Anthropic**	**0.811**	**+0.228**	1.00	0.41
Andrej Karpathy	0.614	**nanoGPT**	**0.707**	**+0.093**	0.05	0.76
Jiayi Weng	0.331	**Tianshou**	**0.420**	**+0.089**	0.25	0.00
Tri Dao	0.532	**FlashAttention**	**0.607**	**+0.075**	0.74	0.54
Lilian Weng	0.590	**lilianweng.github.io**	**0.614**	**+0.024**	0.97	0.97
Harrison Chase	0.479	**LangChain**	**0.502**	**+0.024**	1.00	0.22
Aman Sanger	0.289	**Cursor**	**0.305**	**+0.016**	1.00	0.00
Demis Hassabis	0.664	Google DeepMind	0.490	−0.174	1.00	0.62
Mira Murati	0.442	Thinking Machines Lab	0.222	−0.220	0.97	1.00
Aravind Srinivas	0.663	Perplexity	0.193	−0.470	1.00	0.33

The injection confound in full. Because the injected clause names the artifact and the gold answer also names it, a model can raise its coverage score simply by echoing the injected token, without any additional stored knowledge—the same scoring-leakage channel otherwise closed by withholding contributions from the context (Section 2.2), deliberately opened here. Two features of the result limit this confound without eliminating it: the lift concentrates in low-baseline creators rather than spreading uniformly (pure echo would credit saturated leaders too, and they show none), and the tier gradient tracks recognition deficit (frontier-reasoning +0.008, frontier-chat +0.046, mid-weight +0.106, small +0.095). The clean test—re-judging configuration B against an artifact-redacted gold that credits only facts *not* present in the injected clause—is registered for the next run; pending that, the measured lift is an upper bound.

Table 10. Context-injection experiment. Δ is the creator's NameRank change when the artifact name is added to the disambiguating context (configuration B − configuration A), with the paired t-statistic across the model panel. Bottom block: mean lift by model tier.

Creator	Artifact injected	Δ	paired t
Jiayi Weng	Tianshou	+**0.288**	5.8
Harrison Chase	LangChain	+**0.151**	3.3
Tri Dao	FlashAttention	+**0.094**	2.5
Simon Willison	Datasette	+**0.073**	2.8
Aman Sanger	Cursor	+0.042	1.8
Lilian Weng	lilianweng.github.io	+0.029	0.8
Demis Hassabis	Google DeepMind	+0.023	0.7
Andrej Karpathy	nanoGPT	+0.018	0.7
Mira Murati	Thinking Machines Lab	−0.014	−0.3
Aravind Srinivas	Perplexity	−0.022	−0.9
Dario Amodei	Anthropic	−0.047	−2.2
Mean over pairs		+0.058	
Lift by model tier	frontier-reasoning	+0.008	
	frontier-chat	+0.046	
	mid-weight	+0.106	
	small	+0.095	

Table 11. The Tianshou case: Jiayi Weng against eight OpenAI individual-contributor peers who share the affiliation but not a named artifact. Refusal rate is the fraction of the panel answering "unknown".

Entity	NameRank	Refusal rate
Jiayi Weng	**0.331**	24%
Tianshou (his artifact)	0.420	3%
Trevor Blackwell	0.431	8%
Vicki Cheung	0.300	24%
Pamela Vagata	0.184	32%
Adam Fry	0.084	51%
AJ Ostrow	0.078	60%
Andrea Vallone	0.076	57%
Alex Wei	0.039	51%
Aaditya Singh	0.027	68%
Peer mean	0.152	—
Jiayi Weng within peers	$+1.24\sigma$	—

D.2 The Tianshou case

For each of the 11 verified (creator, artifact) pairs, the full per-model breakdown of $C \to A$ and $A \to C$ (fraction of non-refusal model responses that name the counterpart) is released as a supplementary CSV.

Per-model mention table for Jiayi Weng / Tianshou. Of the 28 non-refusal responses to "tell me about Jiayi Weng," 7 name Tianshou: claude-opus-4.6-think, claude-sonnet-4.6-think, gemini-3-flash-think, gemini-3.1-pro, glm-5.1-think, gpt-5.5-think, kimi-k2.6-think. All other models that respond do not name Tianshou. The 7 mentioning models have judge scores ranging 0.70 to 0.80 (mean 0.71); the 21 non-mentioning responding models range 0.00 to 0.50 (mean 0.35). The score lift from artifact-mention is +0.36.

E External Validity Regressions

Full regression results for NameRank on external impact metrics, restricted to the OpenAlex long-tail researcher cohort ($n = 771$, citations 500–30,000, h-index from OpenAlex).

Predictor set	Full-sample R^2	10-fold CV R^2 (mean $\pm$ sd)
log(citations)	0.137	0.12 ± 0.07
log(h-index)	0.398	0.38 ± 0.09
log(citations) + log(h-index)	0.398	0.38 ± 0.09

Cross-validation is repeated 10-fold (20 repeats, 200 folds, fixed seed); the $\pm$ is the across-fold standard deviation. We use repeated CV rather than a single 70/30 split because at $n = 771$ a single split's held-out R^2 varies by $\sim$0.15 across seeds (one favorable split gave 0.50), which would overstate generalization. The marginal R^2 of log(citations) given log(h-index) is 0.000; the coefficient on log(citations) in joint regression is -0.002, indistinguishable from zero, and citations add no held-out signal.

Decile breakdown of NameRank by h-index (Table 12):

Table 12. Mean NameRank by h-index decile within OpenAlex long-tail researchers.

Decile	n	Mean h	Mean NameRank
D1	77	2.9	0.219
D2	77	8.1	0.329
D3	77	12.4	0.392
D4	77	16.7	0.448
D5	77	20.9	0.464
D6	77	25.8	0.524
D7	77	30.6	0.539
D8	77	36.9	0.574
D9	77	45.4	0.546
D10	77	60.4	0.601

The relationship is near-monotonic (a single dip at D9) and saturates above $h \geq 40$. The bottom decile ($h < 5$) shows NameRank floored at 0.22—the working-researcher floor, not zero.

F Bibliometric Predictor Decomposition

Six bibliometric predictors regressed on NameRank for the $n = 771$ OpenAlex long-tail cohort (Section 3.4), testing whether h-index dominates because it discounts per-paper name attribution. `fractional_citations` divides each paper's citations by its author count; `first_last_citations` counts only papers where the researcher is first or last author; `n_works` is the raw count of works; `mean_authors` is the mean author count per paper. All R^2 on $\log(1 + x)$-transformed predictors.

Table 13. Sole-predictor R^2 on NameRank and marginal R^2 when added to h-index alone. Attribution-weighted measures (fractional, first/last) do not approach h-index, and add almost nothing on top of it; author count carries no signal. The count of distinct works (n_{works}) is the closest cousin of h-index.

Predictor	Sole R^2	ΔR^2 added to h-index
log(h-index)	**0.398**	—
log(n_{works})	0.337	+0.0003
log(first/last-author citations)	0.268	+0.0268
log(fractional citations)	0.154	+0.0026
log(raw citations)	0.137	+0.0001
log(mean authors/paper)	0.0004	+0.0008

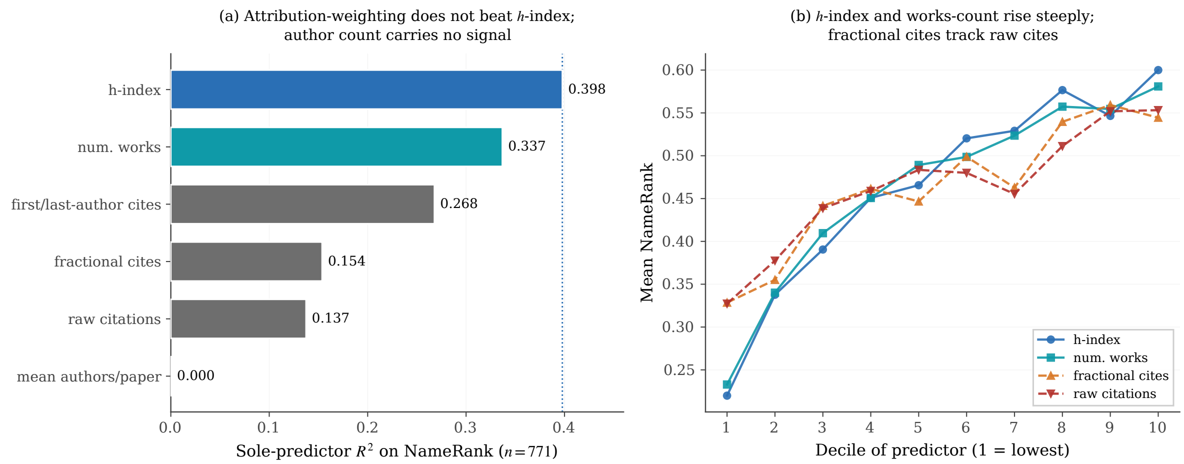

Figure 10. Bibliometric predictor decomposition ($n = 771$). **(a)** Sole-predictor R^2 on NameRank: if h-index dominated by discounting per-paper attribution, the attribution-weighted measures (fractional and first/last-author citations) would match it; instead they barely beat raw citations, and mean authors-per-paper carries no signal. The count of distinct works (n_{works}) is the closest cousin of h-index. **(b)** Mean NameRank by predictor decile: h-index and n_{works} rise steeply and near-monotonically; fractional citations track raw citations on a flat slope.

In a joint OLS over all six (total $R^2 = 0.428$), the standardized coefficients are dominated by log(h-index) (0.579), with first/last-author citations (0.202) a distant second and raw citations *negative* (-0.143); fractional citations (0.109) and mean authors (0.060) are minor. The mechanism is name-recurrence across distinct works, not attribution-per-paper weighting: if the latter drove the result, fractional and first/last-author citations would match or exceed h-index, and they do not.

G Institution and Country Gradient Detail

Table 14 gives the institution-level view behind the Stanford–Tsinghua contrast; Table 15 the full country-level aggregation behind Figure 6.

The well-sampled backbone of the gradient is USA (mean 0.460, 95% CI [0.443, 0.476]) against China (0.311, [0.271, 0.350]) and India (0.292, [0.234, 0.351]): the intervals do not overlap. The apparent leaders (Israel, Singapore, Sweden) sit above the USA in point estimate with intervals that overlap it, and are read as suggestive only.

Table 14. CS-faculty NameRank by institution (selected institutions with $n \geq 4$). Stanford-affiliated faculty score roughly twice Tsinghua-affiliated. The Chinese-institution samples are small ($n = 4$ each); the gradient they anchor recurs at country level (Figure 6), where the samples are larger.

Institution	n	Mean	Min	Max
Stanford University	19	**0.537**	0.14	0.62
University of Washington	26	0.529	0.13	0.68
Cornell University	33	0.484	0.16	0.67
Carnegie Mellon University	65	0.484	0.09	0.66
HKUST	5	0.462	0.24	0.58
Georgia Institute of Technology	6	0.462	0.23	0.65
Johns Hopkins University	4	0.453	0.19	0.67
TU Delft	5	0.445	0.30	0.58
Princeton University	25	0.443	0.18	0.61
University of Michigan	31	0.429	0.16	0.67
Universidade de Lisboa	5	0.427	0.26	0.52
Monash University	6	0.386	0.27	0.65
USTC	4	0.310	0.11	0.45
Peking University	4	0.290	0.22	0.34
Tsinghua University	4	**0.258**	0.15	0.35

Table 15. CS-faculty NameRank by country of institutional affiliation, for countries with ≥ 3 sampled faculty. An *Other/Unknown* bucket ($n = 236$; CS-Rankings affiliations not matched by the keyword-prefix country lookup) is omitted. Only the USA ($n = 302$), China (30), UK (16), Australia and Hong Kong (11), and Brazil and India (10) reach $n \geq 10$; the remaining rows are point estimates with wide intervals and should not be finely ranked against one another.

Country	n	Mean	Median	frac ≥ 0.5
Israel	7	0.506	0.528	57%
Singapore	7	0.497	0.480	43%
Sweden	5	0.492	0.490	40%
Germany	9	0.490	0.488	44%
Switzerland	4	0.489	0.467	50%
Canada	9	0.478	0.463	22%
USA	302	0.460	0.488	46%
Netherlands	9	0.447	0.432	33%
UK	16	0.438	0.440	38%
Hong Kong	11	0.431	0.452	36%
Portugal	5	0.427	0.442	20%
Brazil	10	0.388	0.409	10%
Australia	11	0.341	0.291	18%
France	3	0.312	0.341	0%
China	30	**0.311**	0.285	7%
Spain	8	0.303	0.288	13%
India	10	0.292	0.279	0%

H Cross-Language and East–West Detail

Sub-run cohort selection. The Chinese-prompt sub-run covers 240 entities: the full diagnostic reference set (37), all NOI China gold medalists (29), all DeepSeek-V3 paper authors (69), 30 random samples each from the CMO, CPhO, and MSRA Fellowship cohorts, and 5 each from the writer, filmmaker, and politician cohorts as Western controls—$240 \times 37 = 8{,}880$ records.

Table 16. Cross-language NameRank by cohort: $\Delta = \mathrm{NR}_{\mathrm{zh}} - \mathrm{NR}_{\mathrm{en}}$ on the 240-entity Chinese-prompt sub-run. Chinese prompts lift Chinese-named cohorts and slightly suppress English-coded ones; per-entity extremes in Appendix H.

Cohort	n	$\mathrm{NR}_{\mathrm{en}}$	$\mathrm{NR}_{\mathrm{zh}}$	Δ
DeepSeek-V3 authors	69	0.178	**0.247**	$+$**0.069**
Filmmakers (control)	5	0.765	0.802	$+0.037$
NOI China gold	29	0.368	0.392	$+0.025$
CMO China gold	30	0.301	0.315	$+0.013$
CPhO China first prize	30	0.184	0.194	$+0.010$
Politicians (control)	5	0.361	0.364	$+0.003$
Writers (control)	5	0.650	0.631	-0.020
MSRA PhD Fellowship	30	0.339	0.319	-0.021
Diagnostic reference set	37	0.452	0.422	$-$**0.030**

English-prompt East–West extremes. On English prompts, the strongest Chinese-panel lifts over the Western panel are Chinese-content entities: Wei Dongyi (Chinese mathematics figure; Chinese-panel mean 0.80 vs. Western 0.53) and Zhang Fang (NOI gold; 0.45 vs. 0.20). The strongest Western-panel lifts are non-Chinese names on which Chinese models refuse: Saad Albawi (0.59 Western vs. 0.16 Chinese) and Qingzi Vera Liao (Chinese-name CS faculty at a Western institution; 0.50 vs. 0.15).

Per-model prompt-language deltas. The Chinese-prompt shift is universal rather than vendor-specific: models gaining on Chinese prompts include Western (claude-sonnet-4.6-think $+0.17$, gemini-3.1-pro $+0.12$, gemma-3-12b $+0.12$) and Chinese (deepseek-v4-pro-think $+0.14$, qwen3-235b $+0.10$) models, and models losing include both classes as well (llama-4-maverick -0.20, gpt-5.4 -0.12, glm-4-32b -0.05, kimi-k2 -0.05). The class averages are $+0.014$ (Western) and $+0.028$ (Chinese): both lift slightly, and the dominant signal is entity-name match, not model origin.

Per-entity deltas. The full per-entity $\Delta(\mathrm{NR}_{\mathrm{zh}} - \mathrm{NR}_{\mathrm{en}})$ table is released as a supplementary CSV. Selected diagnostic entries (the largest suppressions beyond the table: Leng Junsheng -0.17, Du Zhuofan -0.15, Aleksander Madry -0.12):

Entity	NR_{en}	NR_{zh}	Δ	Cohort
Jiayi Weng	0.331	0.327	-0.004	reference set
Tianshou	0.420	0.399	-0.021	reference set
tuixue.online	0.250	0.262	$+0.012$	reference set
Andrej Karpathy	0.614	0.549	-0.066	reference set
Tri Dao	0.532	0.488	-0.044	reference set
FlashAttention	0.607	0.470	-0.137	reference set
Bojie Li	0.090	0.105	$+0.014$	reference set
Wang Haoyu (Wang Hao-yu)	0.260	0.430	$+0.170$	CMO China gold
Yao Xuanrong (Yao Xuan-rong)	0.250	0.420	$+0.160$	NOI China gold
Wang Zhongjian (Wang Zhong-jian)	0.260	0.420	$+0.160$	CMO China gold
Zijia Zhu	0.100	0.290	$+0.190$	DeepSeek-V3 author
Peng Zhang	0.160	0.330	$+0.170$	DeepSeek-V3 author

The pattern is consistent: Chinese-character names lift on Chinese prompts; English-coded artifacts lose on Chinese prompts; flat-cross-language entities have name attribution roughly equal in both training corpora.

I News-Event Cohort: Construction and Detail

This appendix documents the construction of the 258-event cohort of Section 3.5 and reports the full tables behind its findings. All construction scripts, inputs, and derived data are released under `experiments/t4_1_news_events/` in the repository.

I.1 Sampling frame and filters

Candidate events are harvested from the "Events" sections of the English Wikipedia year articles (*2021*, *2022*, *2023*) and of 32 country-year articles ("*2021 in India*", "*2022 in Nigeria*", …) chosen to balance nine world regions (2,245 unique linked articles). Each candidate is classified by Gemini 3 Flash Preview (the study's judge family; temperature 0, structured output) from its title, short description, and first paragraph into: discrete event or not, start year and month, primary country, region, and category. An article qualifies as a *discrete event* if it describes a specific occurrence or tightly bounded series (an election, a storm, a trial, a tournament, an accident, a protest wave) that began at an identifiable date; people, organizations, products, laws, ongoing multi-year topics, and list/timeline articles are excluded (1,041 qualify). Post-classification filters require a standard-type article with an intro of ≥ 40 words and a first-year pageview record of $\geq 1{,}000$ total views over ≥ 200 recorded days; 655 events survive. Stratifying by $\log_{10}$(total views) band ($< 10^4$ through $\geq 10^7$) crossed with eight region groups, capped at 14 events per cell, yields the released cohort of 258.

I.2 Attention measures and cross-checks

For each event we retrieve the daily en-Wikipedia pageview series of its article (Wikimedia REST API, user traffic only) for the 365 days from the first day of the event's start month, and define **total attention** (summed daily views), **peak attention** (maximum daily views), and **effective duration** = total/peak in days, linked by the identity $\log(\text{total}) = \log(\text{peak}) + \log(\text{duration})$. Wikipedia pageviews are a standard collective-attention proxy [Mestyán et al., 2013, Ripberger, 2011]; we prefer them to Google Trends because the pageview record is absolute, archived, and reproducible, whereas the public Trends interface returns only normalized indices under heavy rate limits. Two caveats. First, the ledger is attention *to the event's English Wikipedia article*, so all matched-attention comparisons condition on English-audience interest; this is a feature for the region analysis (it holds English exposure fixed) but means globally quiet, locally enormous events enter as quiet. Second, attention can fragment across sibling articles (*Killing of Tyre Nichols* vs. *Tyre Nichols protests*), adding measurement noise that biases dose–response estimates toward zero. A GDELT news-coverage-volume cross-check [Leetaru and Schrodt, 2013] on the event's name phrase over the same window is reported under Full Results below and released with the data.

I.3 Probes and panel

Golds are the article's intro truncated to 100–200 words at a sentence boundary (the main run's Wikipedia gold recipe); disambiguating contexts are generated deterministically from the classification fields only—category phrase, location, month, year—never from article content, mirroring the main run's role/affiliation/field rule. Probe template, judge model, judge prompt, refusal detection, and scoring are unchanged from Section 2.

Panel. Nine of the 37 main-run models could not complete the event run (2026-07): eight lost all OpenRouter routing to provider delistings and deprecations (Ernie-4.5-300b, Mistral-Large-2411, Mistral-Medium-3.1, Ministral-3b, Kimi-K2, Grok-4, GLM-4-32b, Llama-3.2-1b), and Grok-4.20-think—initially recovered via the direct xAI API—lost access when that account exhausted its credits at 143 of 258 events, so its partial coverage is discarded (a model must cover the full cohort to enter the panel mean). The eight delisted models are replaced by the representative newly released models evaluated in the IKP revision [Li, 2026]: Claude Fable 5, GLM-5.2, Qwen3.7-Max, Kimi-K2.7-code, MiniMax-M3, Step-3.7-Flash, Nemotron-3-Ultra-550b, and Nemotron-3-Nano-30b (one representative per newly covered vendor family, plus a second NVIDIA entry to restore the small-model tier lost with Ministral-3b and Llama-3.2-1b; all run in thinking mode per the panel policy). Event NameRank is therefore a 36-model panel mean (28 survivors + 8 replacements) over 9,288 records. Neither change matters for comparability. Recomputing the full main run on the 28-model surviving sub-panel correlates with the released 37-model scores at $r = 0.993$ per entity (mean shift $+0.011$), so main-run numbers are quoted unadjusted alongside event scores, and cross-run placements use survivor-only event means. And the replacements reproduce the survivors' ordering: per-event means over the 8 new models correlate at $r = 0.905$ with means over the 28 survivors (level shift $+0.091$—the 2026-vintage models are uniformly more knowledgeable about 2021–2023 events), the entity-not-panel property of Appendix K on models released a year apart.

Cutoff coverage. Three panel models have training cutoffs inside the final months of the event window (Mistral-Small-24b, 2023-10; Llama-3.1-8b and Llama-3.3-70b, 2023-12); the eight replacement models are 2026 releases with cutoffs far past the window. Neither exclusion matters: dropping the 22 events that begin in October–December 2023 leaves the joint attention decomposition at peak $+0.045$ / duration -0.007; dropping the three models for all events gives $+0.038$ / -0.009 (vs. $+0.042$ / -0.008 headline).

I.4 Full results

Dose–response. Table 17 gives the decile ladder with the coverage/accuracy channel decomposition. Sole-predictor R^2 on NameRank: log total views 0.095 (repeated 10-fold CV 0.09 ± 0.11), log peak 0.119 (0.11 ± 0.12), log effective duration 0.033 (0.02 ± 0.08), log tail interest (mean daily views in days 300–365) 0.065. Gold length alone explains 0.006; adding it to log total views raises R^2 to 0.168 and the attention slope from $+0.042$ to $+0.063$ per decade (gold density suppresses).

Table 17. NameRank by decile of total first-year pageviews ($n = 258$). Attention converts refusal into accurate specifics: accuracy rises steeply, coverage gently.

Decile	n	Geo. mean views	NameRank	Refusal	(cov, acc)
D1	25	2,415	0.379	0.12	(0.46, 0.61)
D2	26	7,395	0.385	0.11	(0.47, 0.60)
D3	26	16,514	0.405	0.12	(0.46, 0.64)
D4	26	39,348	0.464	0.07	(0.54, 0.67)
D5	26	67,583	0.427	0.06	(0.49, 0.71)
D6	25	133,127	0.513	0.05	(0.57, 0.75)
D7	26	251,374	0.484	0.04	(0.55, 0.78)
D8	26	480,323	0.478	0.01	(0.53, 0.85)
D9	26	1,223,071	0.490	0.01	(0.54, 0.85)
D10	26	3,312,033	0.499	0.01	(0.54, 0.88)

Peak vs. duration. Joint regression of NameRank on the standardized log-components: peak $+0.042$ (HC1 se 0.008), duration -0.008 (se 0.009), joint $R^2 = 0.122$ (CV 0.11 ± 0.12); marginal R^2 of duration given peak $= 0.003$. With recurring-name and gold-length controls: peak $+0.070$ (se 0.012), duration $+0.005$ (se 0.010). Within peak terciles, moving from the shortest to the longest duration tercile moves NameRank by -0.07, $+0.01$, and -0.01 respectively—no duration channel at matched peak.

Recurring-series flag. An event is a series instance if substituting a nearby year (or adjacent ordinal/Roman edition marker) into its title yields another standard Wikipedia article; the detector sweeps $y \pm 1..6$ to cover 4–6-year election cycles. 106 of 258 events flag as recurring; at matched attention the penalty is -0.061 (se 0.015). The flag is conservative—renamed series (*2023 Men's FIH Hockey World Cup* vs. *2018 Men's Hockey World Cup*) escape detection and dilute the estimate.

Region and vintage. Table 18 reports raw and attention-adjusted region contrasts. Event-year effects at matched attention: 2022 vs. 2021: -0.009 (se 0.019); 2023 vs. 2021: -0.049 (se 0.018).

GDELT cross-check. Of the 258 name-phrase queries, the rate-limited GDELT DOC API returned data for 159; 118 had nonzero matched coverage. On those, log GDELT coverage volume correlates with log pageviews at $r = 0.33$—the two ledgers partially agree—but predicts NameRank at only $R^2 = 0.006$. We attribute the gap mostly to query noise: GDELT matches the event's literal name phrase, which misses paraphrase references and truncates awkwardly for long titles, and the returned subset is rate-limit-selected rather than random. We therefore rely on the pageview ledger and report GDELT as a weak consistency check on the attention record, not an independent predictor.

Table 18. Region groups: raw mean NameRank and the adjusted delta vs. the Anglo-America & Oceania baseline after controlling $\log_{10}$ total views (HC1 standard errors). The single Global-classified event (the Queen's Platinum Jubilee Beacons) is omitted.

Region group	n	Raw mean	Adj. Δ (se)
Anglo-America & Oceania	45	0.463	(baseline)
Eastern Europe & Russia	28	0.506	$+0.038$ (0.030)
Middle East & Africa	35	0.480	$+0.009$ (0.028)
Western Europe	44	0.453	-0.013 (0.025)
Latin America	32	0.452	-0.008 (0.026)
East Asia	31	0.425	-0.033 (0.031)
South & Southeast Asia	42	0.397	-0.060 (0.028)

J Self-Reported Recognition: Design and Detail

This appendix gives the design, estimation, and full results behind Section 3.7 (released as `experiments/t5_4_self_report/`).

J.1 Design

Entities. 430 total: 400 real entities sampled from the main run, stratified over six panel-NameRank bands ($[0, .05)$, $[.05, .15)$, $[.15, .30)$, $[.30, .50)$, $[.50, .70)$, $[.70, 1]$ with quotas 40/40/70/90/90/70 and at most 12 per cohort), plus the 30 fictional entities of Appendix O as traps.

Pairs. 3,350 unique pairs in five strata: 2,200 with both entities in the recognized region (panel NameRank ≥ 0.15; half constrained to close gaps $|\Delta| < 0.15$), 500 crossing the recognition boundary (≥ 0.30 vs. < 0.05), 250 with both below 0.15, and 400 trap pairs (each fictional entity against 12 real partners spread over the bands, plus 40 fictional–fictional pairs). A further 335 pairs (10%) are re-presented in swapped order as a position-bias probe, giving 3,685 presentations per model. Presentation order (A/B) is randomized once and shared across models.

Probe. Each pair is shown with the main-run disambiguation contexts, and the model is asked which entity it knows more about—"specific, verifiable facts such as works, roles, achievements, or dates"—answering exactly one of `A`, `B`, `EQUAL` (knows both to similar depth), or `NEITHER` (recognizes neither). The three self-report models are the main-run checkpoints GPT-5.5, Claude Opus 4.6, and Gemini 3.1 Pro at main-run settings, so each model's verdicts can be compared against its *own* released per-record behavioral scores. All 11,055 calls returned a parseable verdict.

Estimation. Each model's `A/B/EQUAL` verdicts on real–real pairs are fit with Davidson's tie-extended Bradley–Terry model [Bradley and Terry, 1952, Davidson, 1970]—the batch analogue of the Chatbot Arena aggregation [Chiang et al., 2024]—by ridge-regularized maximum likelihood; `NEITHER` verdicts are treated as abstentions. Confidence intervals are 95% normal-approximation pair-bootstrap (200 refits). Entity-level statistics use entities with at least five decided comparisons. Per-model labels for the binary analyses come from the model's own main-run record: *known* $=$ score ≥ 0.15 and no refusal; *unknown* $=$ refusal or score < 0.05.

Table 19. Self-reported recognition per model. ρ_{panel}: Spearman correlation of the Bradley–Terry self-ranking with panel NameRank over real entities (bootstrap 95% CI). Dir. acc.: fraction of decided verdicts agreeing with the panel ordering on both-known pairs with ΔNameRank ≥ 0.15. Boundary: known-vs-unknown pairs by the model's own labels—share picking the known side / the unknown side / abstaining. Trap: share of fictional-vs-known pairs where the fictional side was preferred (ties included). Reversal: verdict consistency under A/B swap.

Model	ρ_{panel}	Dir. acc.	Boundary (pick/err/abstain)	Trap	Reversal
GPT-5.5	0.61 [.59, .63]	0.82	0.74 / 0.12 / 0.14	1/232	0.88
Claude Opus 4.6	0.53 [.50, .56]	0.81	0.63 / 0.14 / 0.23	0/241	0.85
Gemini 3.1 Pro	0.57 [.55, .59]	0.81	0.72 / 0.09 / 0.16	0/211	0.90

J.2 Results

Partial reconstruction of the panel ordering. Table 19: the self-rankings correlate $\rho = 0.53$–0.61 with panel NameRank and agree with the panel ordering on 81% of decided both-known pairs at gaps ≥ 0.15. Position bias is negligible (reversal consistency 0.85–0.90; first-position win rate 0.46–0.49).

What self-report reads. Panel NameRank factors into a *prevalence* component (the share of the 37 models that know the entity, i.e. score ≥ 0.15) and a *conditional level* (mean score among the models that know it). The self-rankings load on prevalence ($\rho = 0.72$–0.76) and only weakly on conditional level ($\rho = 0.11$–0.19); the models' own behavioral scores do the reverse (conditional level $\rho = 0.60$–0.71, prevalence $\rho = 0.07$–0.15 among known entities). Consequently the self-ranking is nearly uncorrelated with the model's own scores among entities it knows ($\rho = +0.04$ / -0.06 / -0.09), while the three vendors' self-rankings agree pairwise at $\rho = 0.90$–0.91—each model agrees with the *other vendors' introspection* far better than with its *own behavior*. The conditional level is gold-conditional rather than fame-driven (specific dense golds are harder to cover than boilerplate golds; cf. the level-sensitivity in Appendices L and O), which is also why directional accuracy against raw own-score gaps lands below chance (0.34–0.40): raw per-model scores are not a cross-entity depth scale, and we do not use them as one.

Fame-incongruent boundary pairs. Restricting known-vs-unknown pairs to those where corpus fame contradicts the model's own state (the unknown entity has the higher panel NameRank by ≥ 0.05): abstention rises from 10–20% (congruent) to 35–64%, the famous-but-unknown side is claimed in only 0–9% of verdicts, and only GPT-5.5 still sides with its own knowledge on most such pairs (0.65, vs. 0.36 for both others; $n = 26/14/69$, so these splits are indicative). Self-report degrades into abstention, not fame-following fabrication.

Trap arm. Across 684 fictional-vs-known pairs the fictional entity was preferred once (GPT-5.5, 0.4% of its 232; ties included); `NEITHER` was answered on 95–100% of fictional–fictional pairs. Recognition claims are near-perfectly precise; the error budget is spent on under-claiming (abstention on 14–23% of genuine boundary pairs).

Caveats. The behavioral ground truth is one judged response per (entity, model), so response-level noise attenuates all within-model comparisons; the prevalence/conditional decomposition and the panel-side statistics are unaffected. `EQUAL` usage varies by vendor (3% of GPT-5.5 verdicts, 5% Claude, 12% Gemini), which the tie-extended likelihood absorbs. The experiment covers three frontier models; smaller models' self-reports may behave differently.

K Scoring-Rule Validity: Variance, Refusals, and Embeddings

This appendix collects the evidence that the scoring rule—LLM judge, multiplicative coverage×accuracy, panel mean—measures the entity rather than the panel, and that no simpler rule would do (Section 4).

K.1 The score measures the entity, not the panel

Decomposing the total sum of squares over all records (Table 20), the entity factor explains more than three times the variance the model factor does. Model identity contributes a minority share, and the per-model generosity differences that produce it (frontier reasoning models at the generous end, small RL-tuned-against-hallucination models at the strict end) are tabulated in full in the released per-model statistics.

Table 20. Variance decomposition over 211,603 records (5,719 entities × 37 models). Entity and cohort are nested, so their rows overlap rather than partitioning the total; each row is the share of total SS that the factor independently explains.

Source	SS	% of total
Entity	9,417	**35.8%**
Model	2,873	10.9%
Cohort	4,425	16.8%
Residual (interactions)	14,035	53.3%

K.2 Refusal structure validates the multiplicative rule

The refusal rate—the fraction of the panel answering "unknown"—inverts NameRank almost perfectly at cohort level (full cohort table with refusal rates in the released cohort specification): the silent cohorts draw refusal rates above 50%, the universal cohorts near zero. Models honestly refuse on low-recognition cohorts rather than hallucinating, which is the threshold-cleanliness property surfacing in raw response behavior.

The per-model refusal signature shows why the multiplicative rule is load-bearing. The high-refusal cluster is dominated by RL-tuned-against-hallucination models; the zero-refusal cluster consists of *fluent hallucinators* that always produce a confident-looking biography (released per-model statistics). The accuracy axis is what disciplines the latter: a wrong-person bio scores high coverage but near-zero accuracy, so its product contributes nothing. A coverage-only or refusal-only metric would systematically over-credit the fluent-hallucinator cluster.

K.3 Embedding similarity cannot replace the judge

Across 175,397 non-refusal records, the correlation between judge score and embedding similarity is only 0.152, and embedding similarity is essentially flat across judge-score buckets (Table 21). Embeddings measure topical similarity, which is high even for hallucinated bios: a fabricated biography of a CS researcher still embeds close to a gold answer about that researcher's actual subfield. The most extreme disagreements—judge 0.0, embedding ≈0.97—are confident, entirely fabricated bios of Chinese-name long-tail researchers, exactly the fluent-hallucination case the judge exists to catch (a worked example is in the repository documentation). An embedding-only NameRank would credit these as recognized.

Table 21. Mean embedding similarity by judge-score bucket (175,397 non-refusal records). Embedding similarity does not separate unrecognized from recognized entities.

Judge score bucket	n	Mean embedding sim.
0.0–0.1	23,621	0.819
0.1–0.3	14,358	0.798
0.3–0.5	26,291	0.801
0.5–0.7	29,616	0.818
0.7–0.9	52,534	0.843
0.9–1.0	28,977	0.829

K.4 Per-cohort within-entity disagreement

The within-entity model-disagreement (standard deviation across the 37-model panel for each entity, averaged within cohort) ranges over an order of magnitude, from 0.404 (NOI China gold) down to 0.113 (online courses); Figure 11 gives the full ladder.

Interpretation. High within-entity σ cohorts (NOI/ICPC/CMO/IMO/long-tail-OpenAlex) exhibit genuine corpus-pocket-specific recognition: different models trained on different fractions of available corpus see different fractions of the entity's coverage. Low within-entity σ cohorts (online courses, conferences, named methods, books) have universal corpus presence: all models agree because the entity name appears in nearly every training-data source.

The variance pattern is itself a probe of corpus-pocket distribution: cohorts with high σ are the ones where "which model do you ask?" changes the answer.

L Prompt-Paraphrase Sensitivity

The headline metric uses a single fixed probe template (Section 2.1). The *ordering* it produces is robust to wording: panel-mean NameRank correlates at $r = 0.93$–0.98 across template pairs, the cohort ladder is invariant under all four templates—including the working-researcher-above-IMO-gold relationship that underlies the credential treadmill—and template is by far the smallest main effect in the variance decomposition (Table 22). The *absolute level* is not: the original template draws fuller responses, and hence systematically higher coverage, than terser paraphrases, and within a single (entity, model) cell the wording sensitivity is comparable to the model-identity sensitivity. What rescues the metric is the panel mean, whose cross-template variance is more than an order of magnitude smaller than any single model's. NameRank values should therefore be read as template-conditional, and cross-study comparisons must hold the probe wording fixed.

We re-probed a cohort-stratified 200-entity subset (30 each from IMO gold, OpenAlex long-tail researchers, CS faculty, Writers, OSS projects, GPT-5 system-card authors; 10 each from Research papers and Filmmakers) on the full 37-model panel under three paraphrased probe templates (T1–T3) alongside the original (T0), for 29,600 records. The four templates:

```
T0: Tell me what you know about {name}, who is {context}. If you do
```

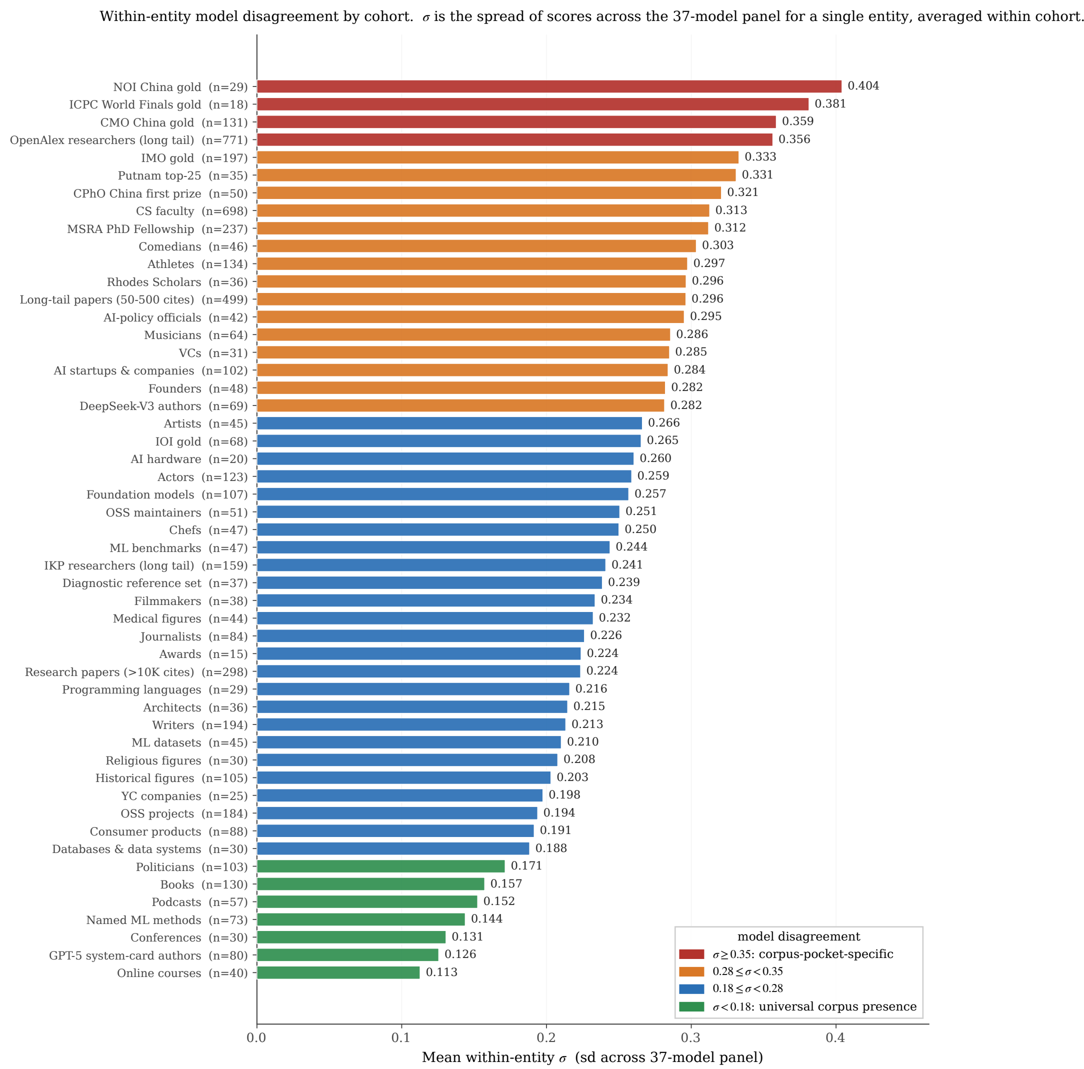

Figure 11. Full within-entity σ ladder across all cohorts with $n \geq 10$. Color bands group cohorts by the four σ regimes (universal $\rightarrow$ corpus-pocket-specific). The high-σ end is dominated by competition-credential and long-tail academic cohorts whose recognition is genuinely corpus-fragment-specific; the low-σ end is dominated by entities whose names appear in nearly every training source (online courses, conferences, named methods, books).

```
    not recognize this entity, respond with "unknown". Limit your
    response to about 150 words.
T1: What can you tell me about {name}? They are {context}. Reply with
    only "unknown" if you are not familiar with this entity. Aim for
    about 150 words.
T2: Please share what you know about {name}, {context}. If unknown to
    you, answer with just "unknown". Keep it under ~150 words.
T3: Briefly describe {name}, who is {context}. Respond "unknown" if
    you have no information. Around 150 words please.
```

Table 22. Additive variance decomposition over the 29,600 prompt-sensitivity records (entity, model, template, cohort as main effects). Template is the smallest main effect. The model share exceeds the full-panel 10.9% (Section 4) because this cohort-stratified subset over-represents silent and universal cohorts, compressing the entity-vs-model contrast; the valid within-experiment comparison is template (1.78%) vs model (26.5%).

Factor	SS	% of total
Entity	985.3	24.19%
Model	1079.9	26.51%
Cohort (entity-implied)	551.9	13.55%
Template	**72.4**	**1.78%**
Residual (after entity+model+template)	1936.3	47.53%

Table 23. Cohort-mean NameRank under each template (sorted by T0). The ordering is preserved; T0 is systematically higher. Pairwise panel-mean Pearson across all six template pairs is 0.927–0.977 (lowest: T0–T3); within-(entity, model) cross-template σ has median 0.109; the panel-mean variance across templates is 0.042× the single-template cross-model variance.

Cohort	T0	T1	T2	T3
Filmmakers	0.761	0.506	0.513	0.532
Writers	0.597	0.388	0.403	0.405
Research papers	0.581	0.391	0.394	0.401
OSS projects	0.504	0.334	0.335	0.331
OpenAlex long-tail researchers	0.465	0.362	0.341	0.455
CS faculty	0.384	0.300	0.295	0.319
IMO gold	0.356	0.290	0.260	0.302
GPT-5 system-card authors	0.047	0.063	0.027	0.070
Panel template mean	**0.420**	**0.305**	**0.294**	**0.329**

M Training-Cutoff Gradient: The Silent Zone Is Intrinsic

A natural worry about the silent zone (Section 2.9) is that it reflects *corpus timing* rather than *intrinsic obscurity*: perhaps low-NameRank entities are merely too recent for current models to have ingested, and a future fleet would lift them. Two contemporaneously-published cohorts test this directly. The DeepSeek-V3 author cohort became nameable in any indexable corpus only when the technical report was posted (December 2024); the GPT-5 system-card author cohort only at the system card's publication (August 2025). The panel's training cutoffs straddle both dates (released with the data; cutoffs span 2023-10 to 2026-03), so a model whose cutoff predates an emergence date *cannot* have ingested the document.

The result (Figure 12) separates two effects. First, a model-vintage confound dominates the cutoff axis: *every* cohort's recognition rises with training cutoff at roughly 0.10–0.20 per year, including cohorts (containing, e.g., Geoffrey Hinton) nameable long before any cutoff—capability and willingness-to-commit drift, not corpus growth. Second, once that drift is netted out by matched difference-in-differences against stable control cohorts, no corpus-timing signal remains: the DeepSeek-V3 cohort's pre/post-emergence jump matches the controls', and the GPT-5 system-card cohort goes *more* silent in post-cutoff models, whose hallucination-averse tuning correctly refuses on a name appearing only in a 486-person contributor list.

Two implications follow. *Recognition is not ingestion*: presence in the training corpus is necessary but far from sufficient, so crossing a cutoff does not manufacture recognition for an entity below the corpus-density threshold. And *NameRank is not comparable across model vintages*: the $\sim$0.13/yr drift means longitudinal claims—including the predictions in Section 5.8—must be made on within-run *relative* gaps, never absolute levels. The thresholded behavior has a mechanistic basis in how transformers store and retrieve facts: feed-forward layers act as key–value memories for factual associations [Geva et al., 2021], with identifiable knowledge neurons per fact [Dai et al., 2022, Meng et al., 2022]; pretrained models function as implicit knowledge bases [Petroni et al., 2019] whose factual capacity scales with model size [Roberts et al., 2020]; accuracy on rare facts scales log-linearly with fact popularity [Kandpal et al., 2023, Mallen et al., 2023]; and popular knowledge actively suppresses less popular knowledge [Zhang et al., 2025]. Below the corpus-density threshold, silence rather than noise is what this retrieval stack predicts—which is what we observe.

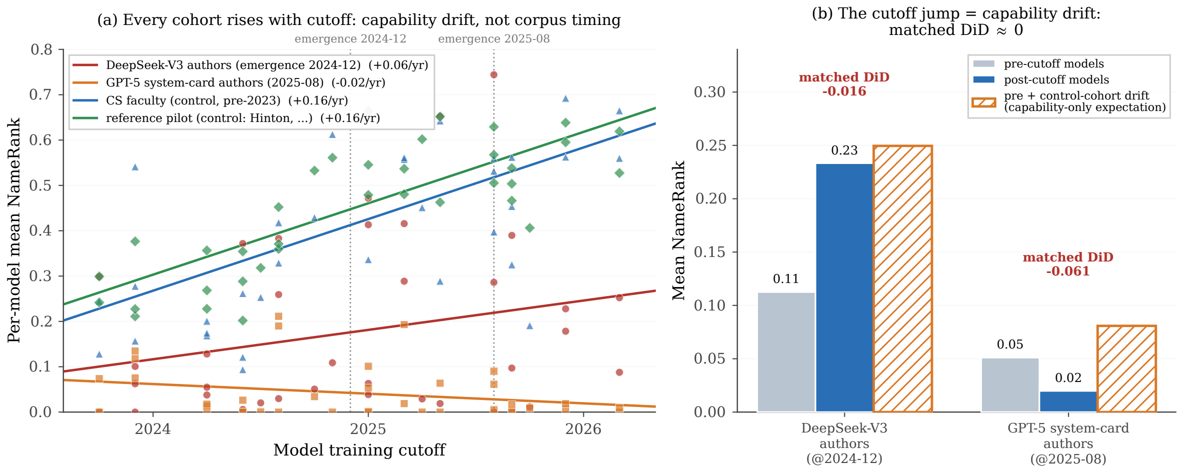

Figure 12. Training-cutoff gradient. **(a)** Per-model mean NameRank vs. training cutoff for two first-appearance "recency" cohorts (DeepSeek-V3 authors, emergence 2024-12; GPT-5 system-card authors, 2025-08) and two stable controls nameable since well before any cutoff. *Every* cohort rises with cutoff at $\sim$0.10–0.20/yr—a model-capability confound, not corpus timing (one control cohort contains Geoffrey Hinton). **(b)** Natural experiment: the recency cohorts' pre/post-emergence jump matches the same-split jump on the stable controls (hatched "capability-only expectation" bar), so the matched difference-in-differences is $\approx$0. Crossing a model's training cutoff is necessary but not sufficient for recognition.

Table 24. Natural experiment: per-model mean recognition split by whether the model's training cutoff predates (pre) or postdates (post) each cohort's emergence date. The raw jump is decomposed by a matched difference-in-differences (DiD) that subtracts the same-split jump averaged over three stable controls (CS faculty, OpenAlex long-tail researchers, IMO gold), which nets out the cross-vintage capability drift. Both matched DiDs are ≤ 0.

Cohort	pre (refusal)	post (refusal)	raw jump	control	DiD
DeepSeek-V3 authors	0.113 (0.38)	0.233 (0.45)	+0.121	+0.137	−**0.016**
GPT-5 system-card authors	0.051 (0.49)	0.020 (0.79)	−0.031	+0.030	−**0.061**

Table 25. Selected per-cohort slopes of per-model mean NameRank on training cutoff (NameRank per year of cutoff), with Pearson r over the 37 models. *Every* established cohort rises with cutoff at ∼0.10–0.20/yr—a model-capability/generosity confound, since these entities were nameable long before any cutoff. The two recency cohorts are *not* steeper; once the drift is removed (Table 24) no corpus-timing signal remains.

Cohort	slope (/yr)	Pearson r
Comedians	+0.196	+0.66
Foundation models	+0.171	+0.76
CS faculty (control)	+0.158	+0.57
Reference set (control; incl. Hinton)	+0.157	+0.77
Research papers	+0.125	+0.74
OpenAlex long-tail researchers (control)	+0.094	+0.32
DeepSeek-V3 authors (recency)	+0.065	+0.23
IMO gold	+0.043	+0.15
GPT-5 system-card authors (recency)	−0.021	−0.25

N Cross-Judge Robustness

NameRank uses a single LLM judge (Section 2.4). To test whether the findings depend on that choice, we re-judged a 615-record stratified sample—spanning the diagnostic reference set, CS faculty, OpenAlex researchers, IMO gold medalists, GPT-5 system-card authors, writers, and OSS projects—with two additional judges from different families (Claude Opus 4.6 and GPT-5.5), holding gold answers and judge prompt fixed.

The three judges agree closely: pairwise Pearson 0.87–0.93 across the full sample (Table 26), 0.95 on the reference-set anchors, and the cohort ladder—including the credential-treadmill ordering—is preserved under all three. A self-preference exists but is small once response capability is controlled (Table 27): Gemini awards its own family a residual +0.062 and GPT-5.5 +0.041, while Claude shows none. This offset shifts per-vendor score tables (released per-model statistics) but not the entity- and cohort-level comparisons that constitute the findings, which average over the full panel and are invariant to the judge. For future runs, Claude is the least self-favoring single judge, and a three-judge mean removes the offset entirely.

Table 26. Pairwise Pearson correlation between judges, by cohort. Agreement is high everywhere except the $n = 15$ OSS projects cell (generic-description disagreement). The Reference set anchors—including its four Chinese-name entities—show the tightest agreement, refuting a "Chinese name → judge disagreement" hypothesis.

Subset	n	Gem–Claude	Gem–GPT-5	Claude–GPT-5
ALL	615	0.868	0.893	0.934
Reference set	185	0.952	0.941	0.972
(Chinese-name subset)	74	0.957	0.945	0.970
OpenAlex long-tail researchers	100	0.855	0.920	0.935
CS faculty	100	0.712	0.894	0.781
IMO gold	100	0.785	0.808	0.924
GPT-5 system-card authors	100	0.781	0.917	0.830
Writers	15	0.872	0.853	0.878
OSS projects	15	0.103	0.455	0.486

Table 27. Capability-controlled in-family lift. For each response, a judge's score is compared to the mean of the *other two* judges on that same response; the lift is the own-family residual minus the other-family residual. Gemini and GPT-5 modestly favor their own family; Claude does not.

Judge	In-family residual	Other-family residual	Lift
Gemini	+0.014	−0.048	+**0.062**
GPT-5	+0.060	+0.019	+**0.041**
Claude	+0.012	+0.022	−**0.010**

Cohort-mean NameRank is stable across judges, and the silent→discriminative→universal ladder is preserved under all three: on this stratified 615-record subsample (whose per-cohort means differ from the full-run means by construction), the GPT-5 system-card cohort scores 0.087/0.165/0.123 under Gemini/Claude/GPT-5, and the OpenAlex long-tail cohort 0.669/0.630/0.631. The headline findings are not a single-judge artifact.

O Synthetic-Null Floor

To establish what NameRank assigns to a name absent from every pretraining corpus, we ran 30 entirely fictional entities—spanning the archetypes the paper measures, with verified-ungoogleable names, cohort-matched contexts, and matched gold answers—through the full pipeline. Table 28 gives the per-archetype floor: near 0.04 for archetypes with dense, specific golds (essentially the main-run silent-zone level), but near 0.20 for the synthetic IMO-gold archetype, whose gold is mostly cohort boilerplate (year, country, medal) that a model can partly satisfy from the disambiguating context alone without recognizing the nonexistent person.

This calibration *sharpens* the credential treadmill rather than threatening it. The olympiad cohorts carry the ∼0.20 boilerplate-gold floor, so IMO gold clears its floor by far less than the OpenAlex baseline clears its ∼0.04 dense-gold floor—the floor-adjusted gap is wider than the raw gap. The corollary caveat: the lowest credential cohorts sit at or below their floor and should be read as silent rather than weakly recognized; differences smaller than the relevant floor are not evidence of recognition.

The floor is concentrated in small open-weights fluent hallucinators: gemma-3-4b (0.293), gemma-3-12b (0.287), and llama-4-maverick (0.259) lead, while 15 of the 37 models score the synthetics at exactly 0 (including grok-4, kimi-k2, qwen3-235b, mistral-large, gemini-3.1-pro, and gpt-5.3, the last two by refusing on 100% of synthetics). Recommended use: treat ~ 0.04 as the noise floor for dense-gold cohorts and ~ 0.20 for short-boilerplate-gold cohorts; differences below the relevant floor are not evidence of recognition.

P Confound Checks: Gold Length, Context, Wikipedia, and Attention

Four alternative explanations for the headline findings were tested and rejected: that the credential gap is a gold-answer-length artifact, that the disambiguating context leaks the answer, that NameRank reduces to Wikipedia presence, and that NameRank reduces to a web-attention ranking.

Gold-answer length. Within-cohort regressions of NameRank on gold-answer length give $R^2 < 0.07$ for IMO (0.067), IOI (0.018), and MSRA (0.006); the high-R^2 cohorts are fame-graded

Table 28. Synthetic-null mean NameRank by archetype. Dense-gold archetypes (founder, CS faculty) floor near the main-run silent level; the short-boilerplate-gold IMO archetype floors much higher, because its generic year+country+medal gold is partly satisfiable from the disambiguating context without recognizing the (nonexistent) person.

Synthetic archetype	n	Mean NameRank	Refusal
IMO gold (boilerplate gold)	6	**0.199**	21%
Chef	1	0.123	27%
Journalist	1	0.092	27%
Podcast	1	0.075	32%
OSS project	4	0.065	39%
Musician	1	0.057	41%
CS faculty	10	**0.027**	45%
Founders	6	**0.012**	47%
All synthetics	30	0.072	—
All except IMO archetype	24	**0.040**	—

mid-tier categories (actor 0.95, writer 0.88) where longer golds are an outcome of fame, not a measurement artifact. The credential-vs-OpenAlex ordering survives a pooled fixed-effects length adjustment (which deepens the gap: pooled slope $b_{\text{chars}} = -1.1\times10^{-3}$) and a within-OpenAlex-slope adjustment. A strict $|\Delta\text{chars}| \leq 50$ matched-pairs test yields $n = 0$ pairs: IMO golds span 359–392 characters, OpenAlex golds 662–789, with no overlap. The closest feasible match (IMO vs. CMO, which do overlap in length) preserves the within-credential ordering.

Disambiguating context. A 288-entity subset re-probed with the specific context replaced by a minimal generic descriptor:

Table 29. Minimal-context ablation. Stripping the specific disambiguating context to a generic descriptor drops absolute NameRank substantially but preserves the cross-entity gradient.

Comparison	Specific context	Minimal context
OpenAlex long-tail researchers (mean)	0.446	0.077
IMO gold (mean)	0.345	0.039
Stanford mean	0.565	0.355
Tsinghua mean	0.258	0.062
Stanford−Tsinghua gap	**0.308**	**0.293** (−4.8%)

Absolute levels fall sharply (the model cannot identify which researcher without the context), but the Stanford–Tsinghua institutional gap shrinks by only $\sim 5\%$: context anchors disambiguation, it does not leak the gold answer.

Wikipedia presence. On the $n = 771$ long-tail cohort, a binary has-Wikipedia flag explains $R^2 = 0.022$ of NameRank (log-sitelinks and log-pageviews add $+0.007$ and 0.000 on top), versus $R^2 = 0.398$ for $\log(h\text{-index})$ alone. Adding $\log(h\text{-index})$ to the full Wikipedia block lifts R^2 from 0.029 to 0.405—a marginal R^2 of 0.376. Across the full 5,719-entity cohort, Wikipedia presence explains $\sim 8\%$ of variance and cohort identity $\sim 47\%$. Wikipedia presence accounts for only $\sim 1/4$ of the Stanford–Tsinghua gap. NameRank tracks bibliometric productivity, not a Wikipedia-yes/no flag.

Web attention. Attention metrics place entities on a single cross-domain axis, so we test the strongest freely available one directly: 24-month en-Wikipedia article pageviews (2023-07 through 2025-06; Wikimedia REST, user traffic) for every entity whose article the strict-disambiguation lookup above resolves—1,617 entities after excluding 48 whose articles were created after the window and 16 disambiguation false positives. Figure 13 summarizes three results. First, the metric is undefined for the 71% of entities with no article, whose NameRank spans 0.16–0.68 (p10–p90)—the entire discriminative zone, where the credential, faculty, and long-tail findings live. Second, where defined, log-pageviews explain $R^2 = 0.06$ of NameRank overall and 0.10 within cohort; on the 58 long-tail researchers where both predictors exist, the pageview correlation is *negative* ($r = -0.33$) while $\log(h\text{-index})$ alone reaches $R^2 = 0.39$. Third, the within-cohort correlation carries a sign structure: positive for celebrity cohorts (actors +0.81, athletes +0.77, writers +0.62), negative for technical ones (OSS projects −0.54, databases −0.44, programming languages −0.35). The divergence is informative rather than noise: Chainer (6.4K views over the window; NameRank 0.94) and Theano (26K; 0.89) are corpus-saturated tools nobody looks up anymore, while Stable Diffusion (1.3M views; 0.38) draws enormous current traffic against a shallower accumulated corpus. Pageviews measure the *flow* of current curiosity; NameRank measures the accumulated *stock* of name-bearing text as absorbed at training time (Section 5.3)—related axes, not the same axis. The excluded post-window articles make the point from the other side: the vLLM and Ollama articles were created only in 2026, yet both entities already score in the discriminative band—recognition no pageview record could have predicted.

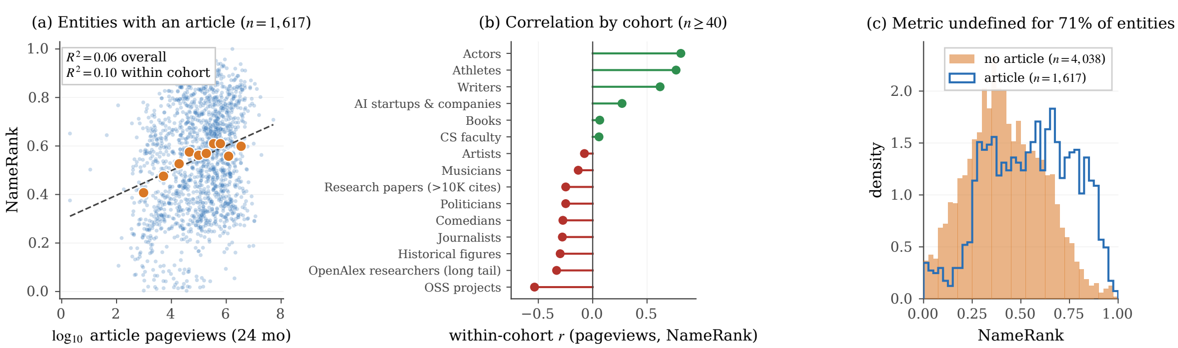

Figure 13. NameRank is not Wikipedia-pageview rank. **(a)** NameRank against $\log_{10}$ 24-month article pageviews for the 1,617 entities with a resolvable article (small dots: entities; large markers: decile means): $R^2 = 0.06$ overall, 0.10 within cohort. **(b)** Within-cohort correlation for cohorts with $n \geq 40$ matched entities: positive only for celebrity cohorts, negative for technical and production cohorts—current attention flow vs. accumulated corpus stock. **(c)** NameRank distribution for entities with and without an article: the attention metric is undefined for the 71% without one, whose scores span the discriminative zone.

Q Gender Gap by Cohort

First-name-inferred gender (via `gender-guesser`; ambiguous and unisex names dropped) for the person cohorts, restricted to cohorts with ≥ 5 man- and woman-coded names. Table 30 reports the man-minus-woman NameRank gap.

The within-researcher-cohort gap (+0.027 to +0.054, ∼6–8% relative attenuation) is roughly 4× smaller than the ∼2× East-Asian-surname attenuation reported by Li [2026], but repro-

Table 30. Man-minus-woman NameRank gap by cohort. The controlled researcher cohorts (top) show a small reproducible man-coded advantage that survives institution- and h-index-matching. The cross-profession cohorts (bottom) show much larger raw gaps that reflect real-world fame differences in the sampled individuals, not a NameRank artifact—hence the bias claim is restricted to the controlled researcher comparison.

Cohort	n_{man}	n_{woman}	Δ (man−woman)	p
CS faculty	354	97	+0.038	0.014
institution-matched (60 pairs)	—	—	+0.054	0.042
OpenAlex long-tail researchers	388	76	+0.031	0.182
h-index-decile-adjusted	—	—	+0.032	—
IKP long-tail researchers	56	22	+0.044	0.015
Actors	38	63	+0.341	$< 10^{-3}$
Athletes	64	32	+0.254	$< 10^{-3}$
Writers	62	77	+0.041	0.41
Comedians	18	17	−0.045	0.23

ducible across three researcher cohorts and robust to dropping medium-confidence gender labels. The large cross-profession gaps (actor, athlete) are excluded from the bias claim because the sampled men and women in those cohorts differ in actual fame.

Name-coding caveat. `gender-guesser` infers gender from first names against a predominantly Western-name dictionary; it returns “unknown” or “andy” (androgynous) for most Chinese, Korean, and other romanized East-Asian given names, which are then dropped. The gendered sub-sample is therefore enriched for Western names, and the residual gender gap is partly entangled with the East-Asian-surname attenuation it sits beside—a name that codes as gender-unknown also tends to be one of the low-recognition non-Anglophone names. The n_{woman} counts (e.g. 76–97 across the researcher cohorts) reflect this attrition. We report the gap as an inherited corpus-and-judge property restricted to the controlled within-cohort comparison, not as a clean estimate of a pure gender effect net of name origin; disentangling the two would require a gender labeling that is accurate on CJK names (e.g. manual coding or a name-origin-aware classifier).

R Limitations Summary Table

Table 31. Compact summary of methodological caveats, their direction (could overstate or understate findings), and mitigations available for future runs.

Caveat	Direction	Mitigation
Single-probe-per-entity; absolute levels are wording-conditional (Appendix L)	Shifts cardinal levels ∼0.10–0.20; rankings unaffected	*Tested:* four-template re-probe shows ordering robust ($r = 0.93$–0.98); report values as T0-conditional and hold wording fixed across studies
Judge family-bias (Appendix N)	Gemini-judging-Gemini lift +0.062 inflates per-vendor score tables; cross-entity findings unaffected	*Tested:* 3-judge re-grade ($n = 615$); report per-vendor tables under Claude or as a 3-judge mean
Cohort labels not fully orthogonal	Overcounts entities in cross-cohort comparisons	De-duplicate by name within long-tail academic categories
Chinese-prompt context fields remained English	Understates cross-language effect	Pure-Chinese prompts for the full 240-entity sub-run
Single judge on the full run (Gemini 3 Flash Preview)	Family-specific scoring style on per-vendor tables	*Tested:* non-Gemini replication (Claude, GPT-5) on $n = 615$; a full multi-judge consensus run would remove the offset
Cross-section, not longitudinal	Cannot infer name-propagation rates	Re-run at each major model-release cycle; release as NameRank v2, v3, ...
Scores not comparable across model vintages (Appendix M)	∼0.13/yr capability drift inflates any cross-fleet level comparison	*Tested:* silent zone is intrinsic, not timing (matched DiD ≈0); make longitudinal claims on within-run relative gaps only
Researcher mobility confound in country gradient	Inflates US averages, deflates China averages	Re-classify by country-of-PhD or country-of-doctorate-training as alternative slicing
Tsinghua/Peking small samples ($n = 4$)	Institution-level reads noisy at the bottom	Expand sample by drawing from CCF rank-A faculty rosters
Tianshou case study sample of 1 for per-response mediation analysis	Cannot generalize the +0.36 score lift cleanly	Extend per-response mediation analysis to all 11 verified (creator, artifact) pairs
Inherited gender bias (Appendix Q)	Man-coded researcher names score +0.03–0.05 over woman-coded	*Measured, not corrected:* report alongside the East-Asian-surname effect; restrict to controlled within-cohort comparisons
Short-boilerplate-gold cohorts have an elevated noise floor (Appendix O)	Olympiad-credential scores up to ∼ 0.20 are partly context-prior leakage	Read scores against the synthetic floor (∼ 0.04 dense-gold, ∼ 0.20 short-gold); the floor widens, not narrows, the credential gap
Institution variance share inflated by category count (Section 3.4.2)	Naive $R^2 = 0.54$ counts 319 dummies on 698 points; true share ∼16%	*Corrected:* report adjusted R^2/ω^2/ICC (∼0.16) and the large-cell Stanford–Tsinghua contrast; country-level aggregation is not inflated
Context-injection lift includes a coverage-echo component (Section 3.3.4)	Overstates the named-artifact-amplification effect; lift is an upper bound	Re-judge configuration B against an artifact-redacted gold crediting only non-injected facts; report the residual lift
Gold/training-source overlap for Wikipedia-derived golds (Section 2)	For ∼60% of entities, coverage partly measures Wikipedia memorization	*Tested:* a binary Wikipedia flag explains only ∼2% of long-tail variance vs. 40% for h-index (Appendix P)
Event attention ledger conditions on English-Wikipedia interest (Appendix I)	Matched-attention region contrasts hold English exposure fixed; globally quiet, locally enormous events enter as quiet	Add non-English pageview ledgers (major Wikipedia editions) as parallel exposure measures
Event attention fragments across sibling articles (Appendix I)	Attenuates the dose–response slope toward zero	Aggregate pageviews over each event's article cluster before computing the ledger
Event panel differs from the main-run panel (36 models: 28 survivors + 8 replacements)	Cross-run level comparisons shift (+0.011 survivors, +0.091 replacements)	*Tested:* survivor sub-panel $r = 0.993$ with released scores; replacements rank events at $r = 0.905$; quote cross-run placements on the survivor sub-panel